Fpt University

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ISP392 PROJECT REPORT

Software Design Specification — Version 3.0

HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM (HSWRMS)

A Wellness Resort & Spa Management System

– Hanoi, Sep 2026 –

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Change Log

*A - Added M - Modified D - Deleted

Date	A* / M / D	In charge	Change Description
02 Nov 2026	M	Not supplied	Final template restoration, source-grounded module diagrams, and evidence-status reconciliation.
03 Nov 2026	M	Not supplied	Source-aligned architecture, UML package and domain ERD views; expanded M1-M6 class diagrams and sequential figure references.

I. High Level Design

1. Software Architecture

HSWRMS covers identity and sensitive health profiles (M1), retreat packages and accommodation booking (M2), spa/therapy scheduling (M3), dietary food service and inventory (M4), folio/invoice/payment/checkout (M5), and system administration/permissions (M6). This SDS documents source-backed design at the repository snapshot and separates it from target decisions and unverified operation.

Phase 0.1-F was a static technical evidence pass: 46 FR records, 152 route candidates, 43 service classes, 48 repository/entity/table records, 88 test classes and 737 annotated test methods were indexed. Nineteen FRs had partial direct controller/dependency candidates, 27 were unmapped, and zero complete FR-to-acceptance chains were established. Zero tests were run in that pass. These are inventory counts, not product-completeness measures.

1.1 Architecture and system scope

Backend source is a Java 17 / Spring Boot 3.3.4 modular monolith using Spring MVC, Spring Security, and Spring Data JPA/Hibernate. The React 19 / TypeScript / Vite client calls REST controllers; module packages are not independently deployed services. Shared security, errors, adapters, repositories, and scheduled work cross package boundaries. SQL Server is the intended database target; H2 appears in test configuration. No production topology, live SQL Server catalog, or provider operation was verified.

Figure 1 shows source layers and marks the SQL Server/provider relationships as targets, not deployed connections.

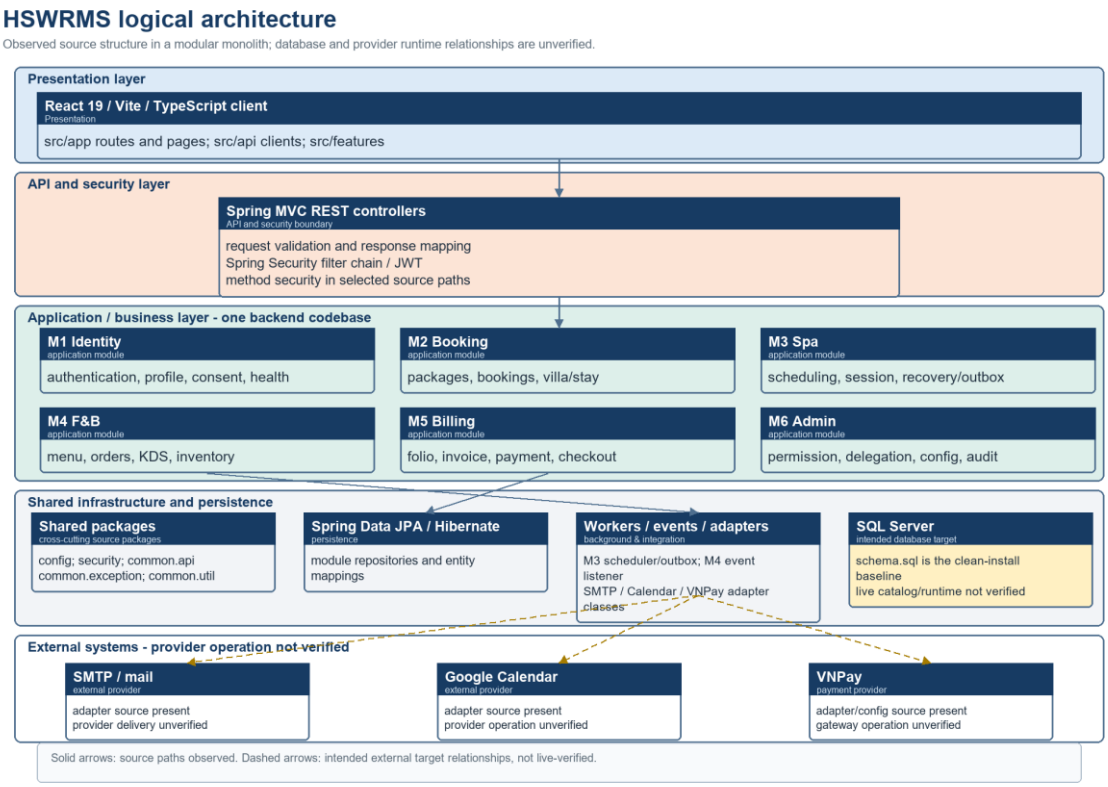


Figure 1. Source-aligned logical architecture. Package layers and six modules are observed in code; SQL Server/provider relationships are targets, not live-verified connections.

Architecture component responsibilities

Component	Responsibility	Main source evidence	Dependency / boundary
React 19 / Vite client	Renders route-oriented screens and calls backend APIs.	src/app, src/api, src/features	UI guards do not prove API authorization or complete screen integration.
Spring MVC and Security	Maps REST requests; JWT filter and selected method guards are source-backed.	config, security, module controllers	Complete route-level access behavior remains unverified.
M1-M6 modules	Own identity, booking, spa, F&B, billing and administration code paths.	m1identity through m6admin	One modular monolith; packages are not independently deployed services.
Shared packages	Provide cross-cutting configuration, security, response, exception and utility types.	config, security, common.api, common.exception, common.util	Shared dependencies do not imply module ownership.
Persistence	Repositories and entity mappings use Spring Data JPA/Hibernate.	module repository/entity packages	SQL Server is the intended target; live catalog and provider behavior are not verified.
Workers, events and adapters	Source includes M3 scheduling/outbox, M4 events and external adapters.	m3spa.scheduler/service; m4fnb.event/integration	Scheduler activation, SMTP, Calendar and VNPay operations are unverified.

Area	Source organization	Boundary interpretation
Frontend	src/app providers/router/guards; src/api and src/features; role-oriented pages	Routes/API clients do not prove operational screen integration. Frontend guards are not authorization.
Shared backend	config, security, common/errors, shared DTOs and ports/adapters	JWT filter and method security are source-backed; full route policy is not.
M1–M6 backend	m1identity, m2booking, m3spa, m4fnb, m5billing, m6admin	Package ownership is technical grouping, not approved data/write ownership.
Persistence	JPA entities/repositories and schema.sql; SQL Server profile	schema.sql is the clean-install baseline; live catalogs/constraints/indexes unverified.
Tests	backend/src/test and frontend test sources	Prior reports are historical evidence, not executed in this task.

2. Package Diagram

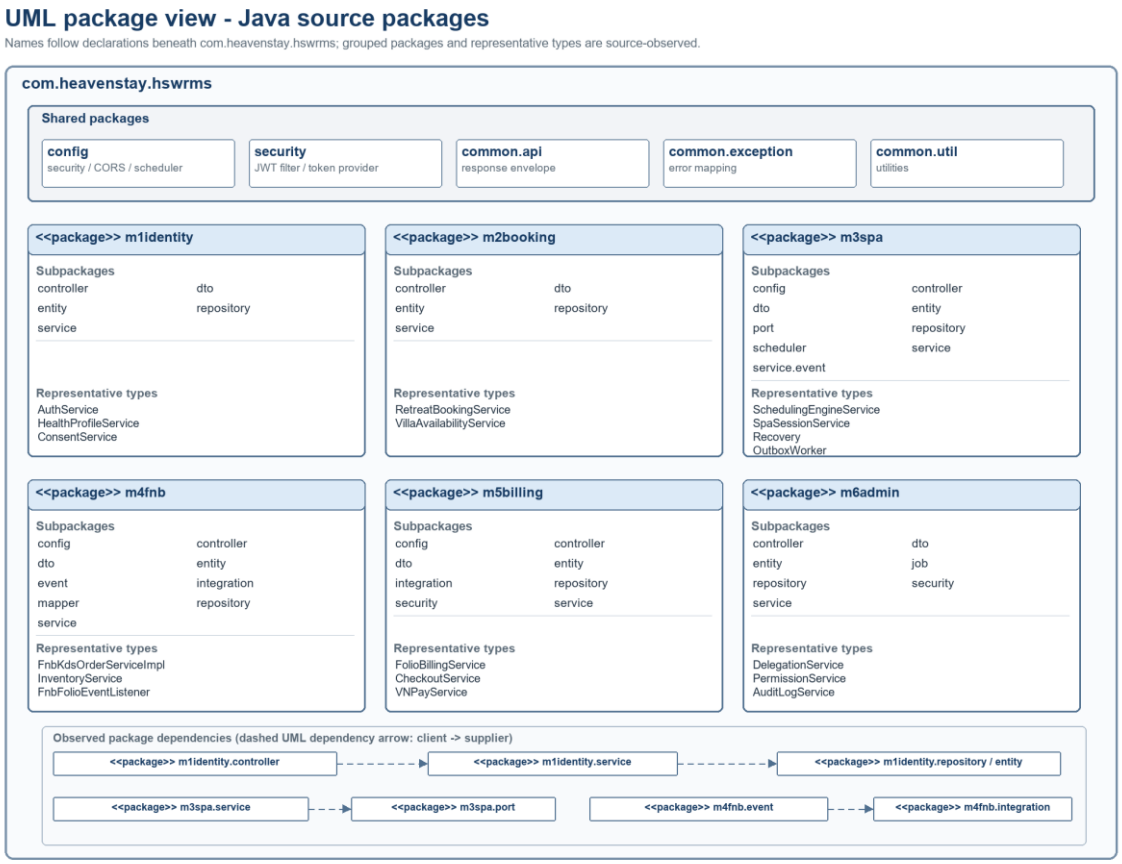


Figure 2. UML package structure beneath com.heavenstay.hswrms. Java package names and grouped subpackages are source-observed; Figure 3 separately shows cross-module calls and context reads.

Package description and representative types

Package	Observed subpackages	Representative source types	Responsibility / boundary
m1identity	controller, dto, entity, repository, service	AuthService; HealthProfileService; ConsentService	Identity, customer profile, consent and health data; module-local persistence.
m2booking	controller, dto, entity, repository, service	RetreatBookingService; VillaAvailabilityService; ReceptionistOperationService	Booking and physical-villa workflow; PaymentTransaction mapping conflict remains.
m3spa	config, controller, dto, entity, port, repository, scheduler, service, service.event	SchedulingEngineService; SpaSessionService; SessionRecoveryService; SpaDispatchOutboxWorker	Scheduling, lifecycle, recovery and provider ports; external delivery is unverified.
m4fnb	config, controller, dto, entity, event, integration, mapper, repository, service	FnbKdsOrderServiceImpl; MealOrderService; InventoryService; FnbFolioEventListener	Menu, order, KDS and inventory; direct + event M5 callers are distinct.
m5billing	config, controller, dto, entity, integration, repository, security, service	FolioBillingService; CheckoutService; VNPayService	Folio, invoice, payment and checkout; provider operation is unverified.
m6admin	controller, dto, entity, job, repository, security, service	PermissionService; DelegationService; BusinessParameterService; AuditLogService	Permission and administration; consumer coverage is incomplete.
shared	config; security; common.api; common.exception; common.util	JwtAuthenticationFilter; shared response/error types	Cross-cutting packages; not separate deployable services.

Within-module source layering includes controller-to-service and service-to-repository/entity dependencies; M3 also has port/scheduler/event packages and M4 has event/integration packages. Cross-module edges are shown separately in Figure 3.

2.1 Module responsibilities and dependencies

Module	Responsibility / technical owner	Observed dependency and limit
M1	Identity, authentication, profile, consent and health projection.	Shared security, M2 customer/booking, M3/M4 minimized data use. Provider runtime and privacy approvals unverified.
M2	Retreat package, booking, villa inventory and reception/check-in.	M1 identity; M3 stay adapter; M5 deposit/payment adjacency. M2 owns Booking/deposit rules by Rev 1.1; M5 owns PaymentTransaction.
M3	Candidate scheduling, reservation, session/confirmation, recovery and outbox.	M1/M2 stay context; M5 charge adapter. No M3→M4 direct path found in reviewed scan; Calendar client absent.
M4	Menu, meal order, KDS, recipe and inventory.	M1 dietary projection; M2 booking; M5 billing; M6 audit. Allergy, location and entitlement runtime remain unverified.
M5	Folio, invoice, payment and checkout code paths.	Reads M1–M4 sources and M6 audit. M5-owned idempotent charge and PaymentTransaction are Rev 1.1 directions, not proof every path enforces them.
M6	Permission catalog, role mapping, delegation, audit and configuration.	Shared backend security and module audit calls. Least-privilege, time-limited, auditable delegation is target direction; complete consumer mapping is pending.

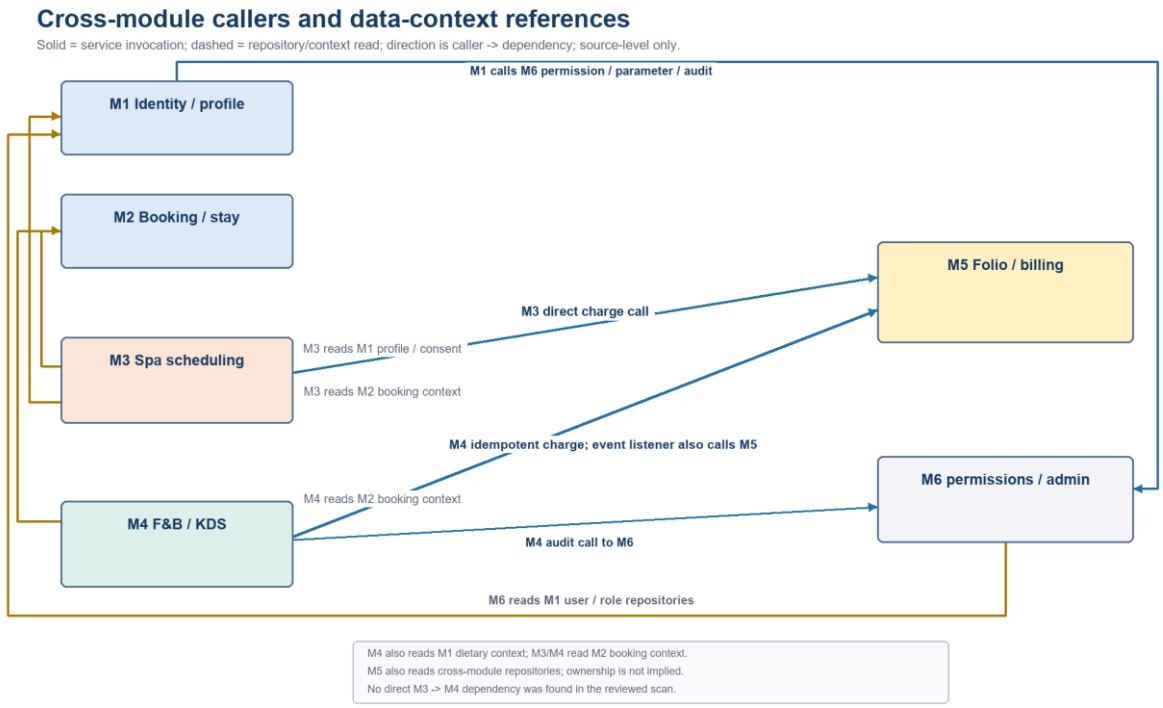


Figure 3. Cross-module caller and data-context dependencies. Solid arrows show observed service calls; dashed arrows show repository/context reads. M1 and M6 depend on one another through distinct calls; edges do not imply ownership or complete runtime contracts.

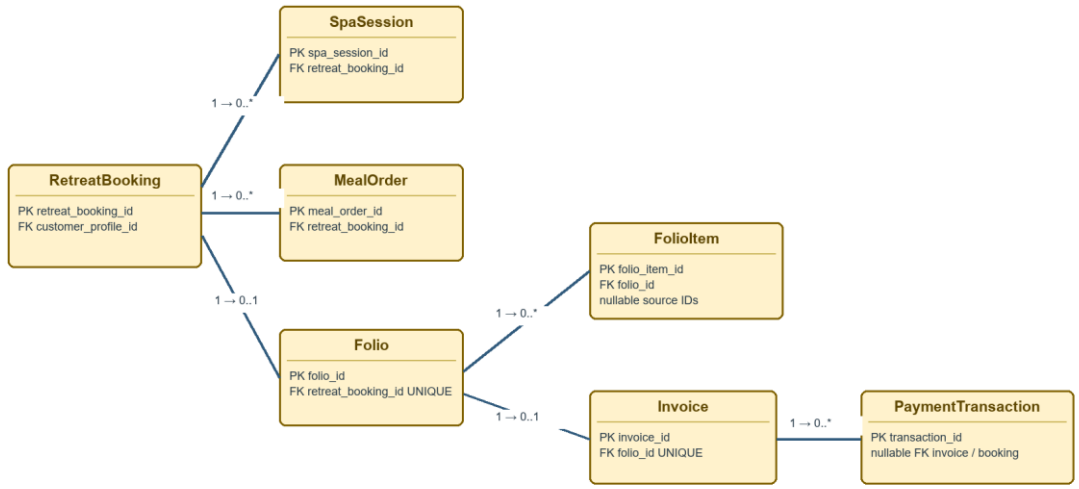
3. Database Design

Static schema review records 52 CREATE TABLE declarations and 52 unique names (50 unconditional and two guarded: SpaScheduleHistory and SpaDispatchOutbox). This describes inspected schema.sql source, not a live catalog or proof that migrations ran. Entity annotations, baseline DDL, migration files, and any deployed database are separate evidence layers.

Domain	Representative entities/tables	Integrity / ownership note
M1	User, AuthIdentity, CustomerProfile, HealthProfile, ConsentRecord, allergy/diet, deletion request	Consent/retention policy and deployed privacy controls need evidence.
M2	RetreatPackage, RetreatBooking, ResortZone, VillaCategory, PhysicalVilla, ResidenceRecord	Villa Category differs from Physical Villa; zone and allocation rules remain.
M3	SpaSession, Therapist, TreatmentRoom, SpaTreatment, SpaScheduleHistory, SpaDispatchOutbox	Resource bundle: time + qualified therapist + compatible room. FK/index/locking state unverified.
M4	MenuItem, Ingredient, MenuItemIngredient, IngredientBatch, MealOrder, MealOrderItem	Usable quantity/expiry/condition are guards; deployed behavior unverified.
M5	Folio, FolioItem, Invoice, PaymentTransaction, adjustments	Folio is mutable; Invoice is a formal snapshot. OI-01 ownership risk remains in source.
M6	Permission, RolePermission, SystemConfiguration, RightDelegation, AuditLog	Canonical contract and operational immutability/retention unverified.

Figure 3 — Selected schema relationships

Static ERD excerpt from schema.sql; explicit FK and UNIQUE/nullability declarations only.
This is not a live SQL Server catalog or complete database ERD.



Legend: solid arrows are schema FOREIGN KEYs; labels show parent-to-child optionality from FK plus UNIQUE/nullability.
Selected excerpt only. PaymentTransaction has nullable booking_id and invoice_id with a payment-type XOR check; deployment state is unverified.

Figure 4. Selected cross-domain relationships declared in schema.sql; excerpt only, not a complete or deployed ERD.

Phase 0.1-F mapped 50 entity classes, 48 repository interfaces, and 110 relationship annotations. Figure 4 is a selected schema-level relationship excerpt, not a complete or deployed ERD. Source annotations or DDL establish only source-level mappings; live foreign keys, trust, cascades, indexes, nullability, SQL types, and SQL Server application remain unverified. See SDS_DATABASE_DICTIONARY_FINAL.md for the detailed static inventory.

Domain ERD views and selected table descriptions

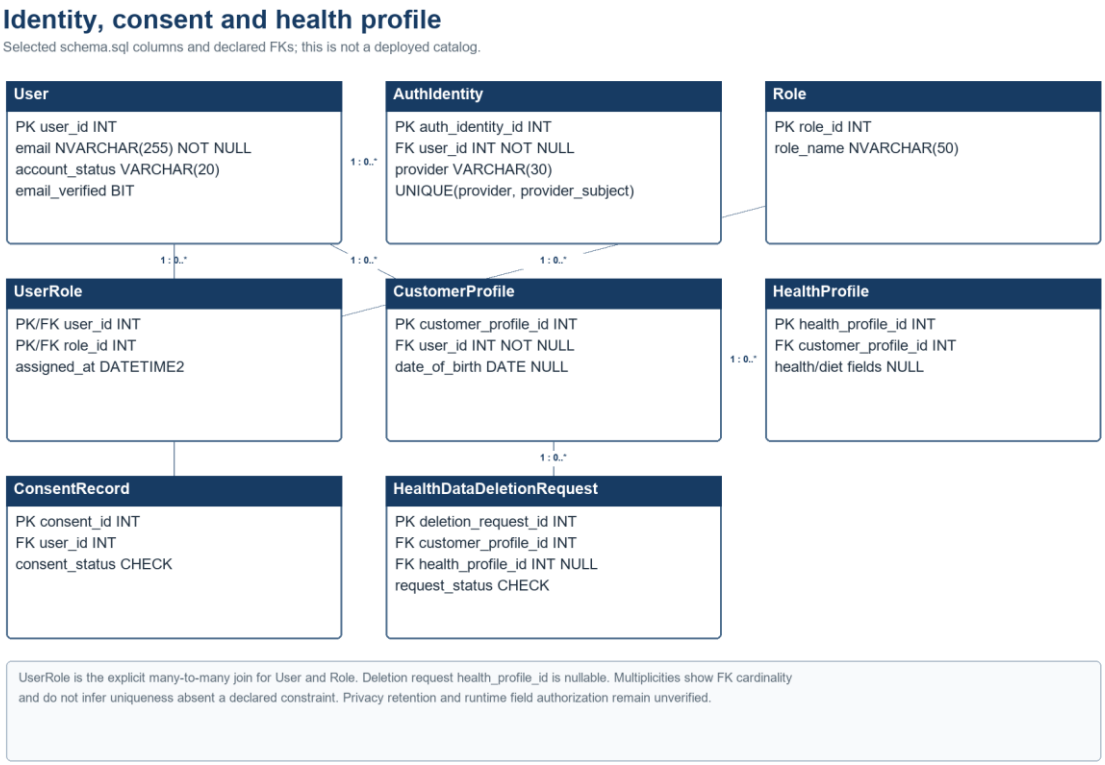


Figure 5. Identity, consent and health-profile ERD. Selected PK/FK fields and nullability are read from schema.sql; runtime catalog and privacy controls are unverified.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

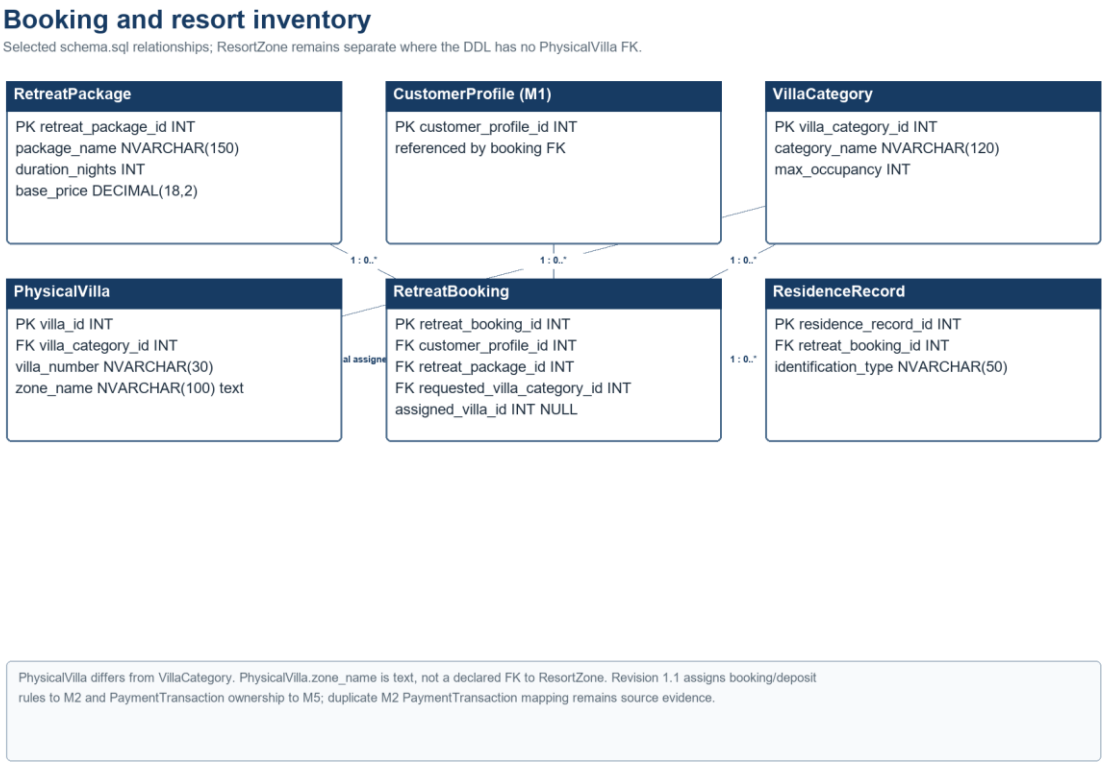


Figure 6. Booking and resort-inventory ERD. RetreatPackage, CustomerProfile and VillaCategory FKs are source-declared; ResortZone is not joined to PhysicalVilla by an inspected FK.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

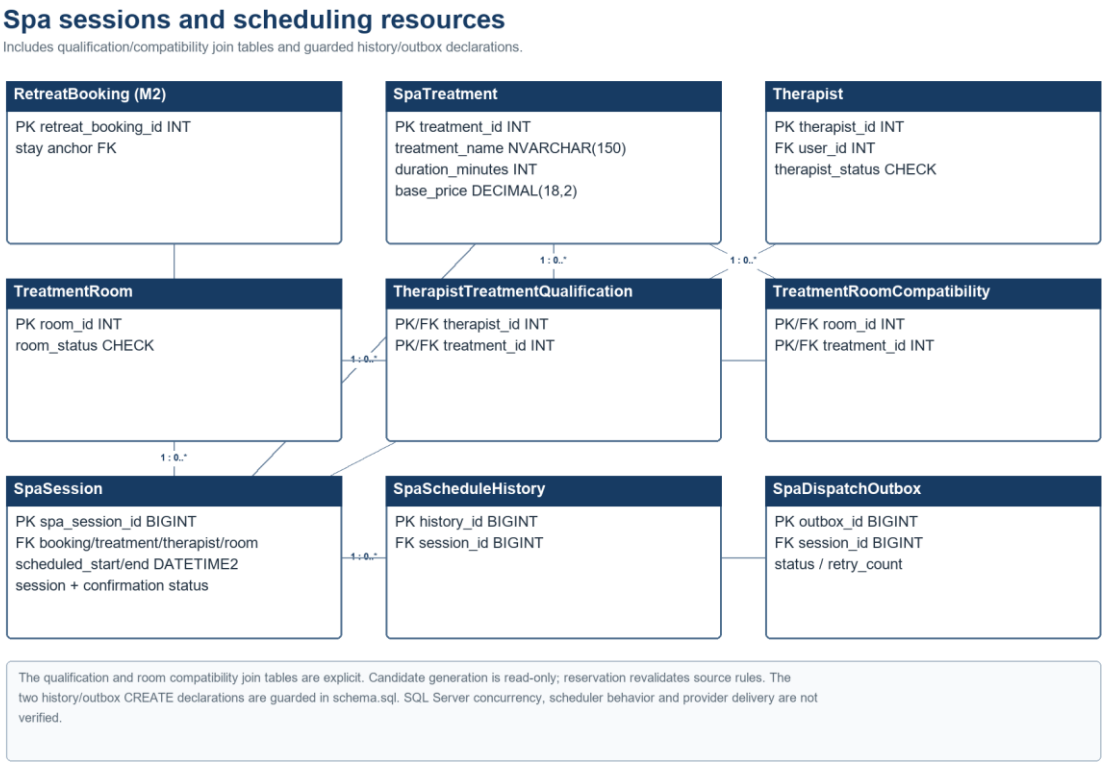


Figure 7. Spa scheduling ERD. Qualification and room compatibility use explicit join tables; history/outbox declarations are guarded in schema.sql.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

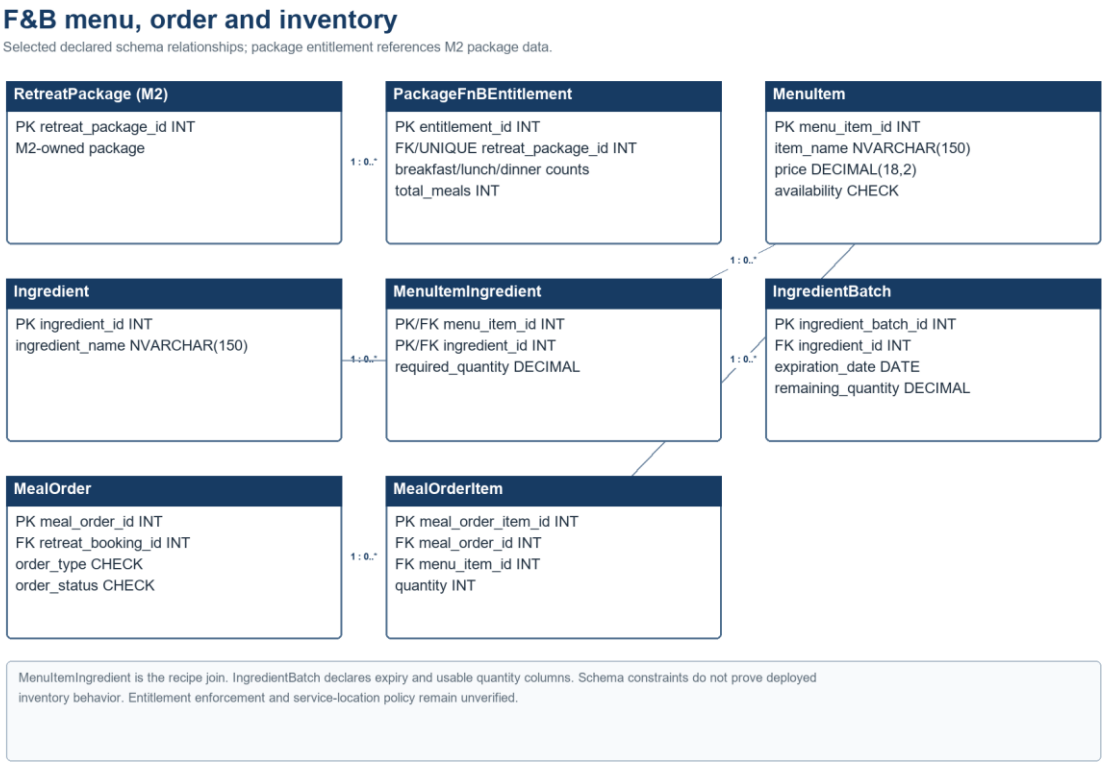


Figure 8. F&B menu, order and inventory ERD. Recipe ingredients and package entitlement constraints are source-declared; deployed enforcement is unverified.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

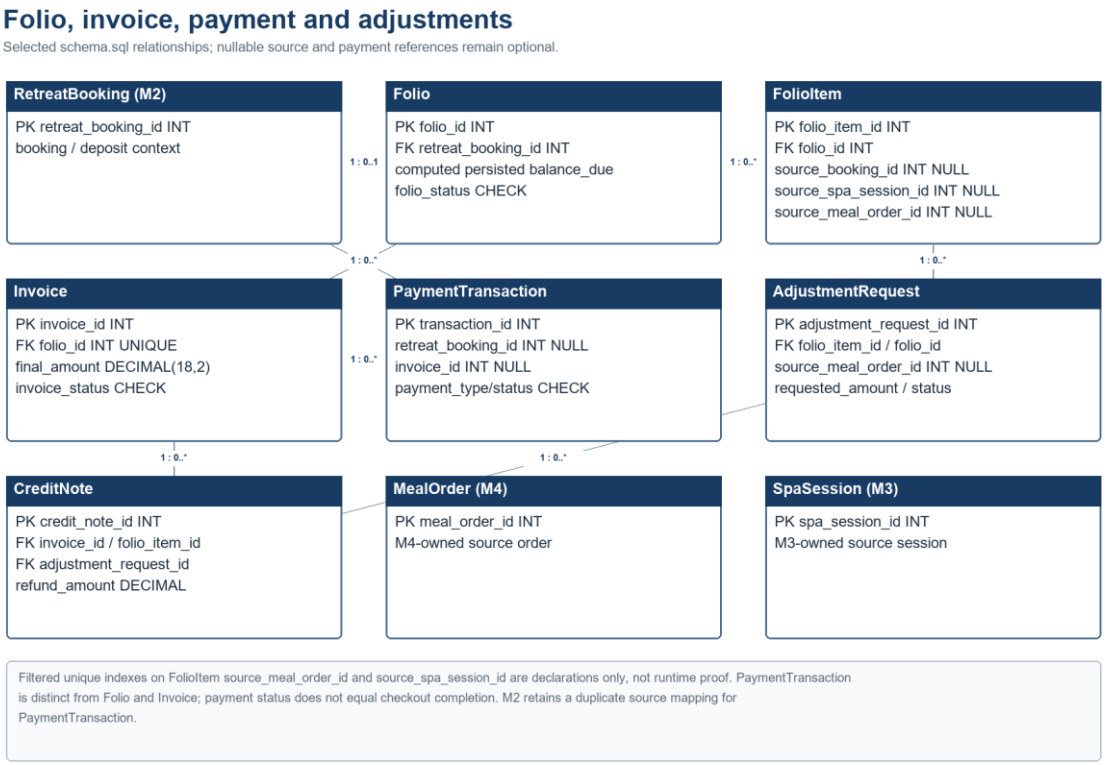


Figure 9. Billing ERD. Folio, FolioItem, Invoice, PaymentTransaction and source references remain separate; filtered unique indexes are schema declarations only.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

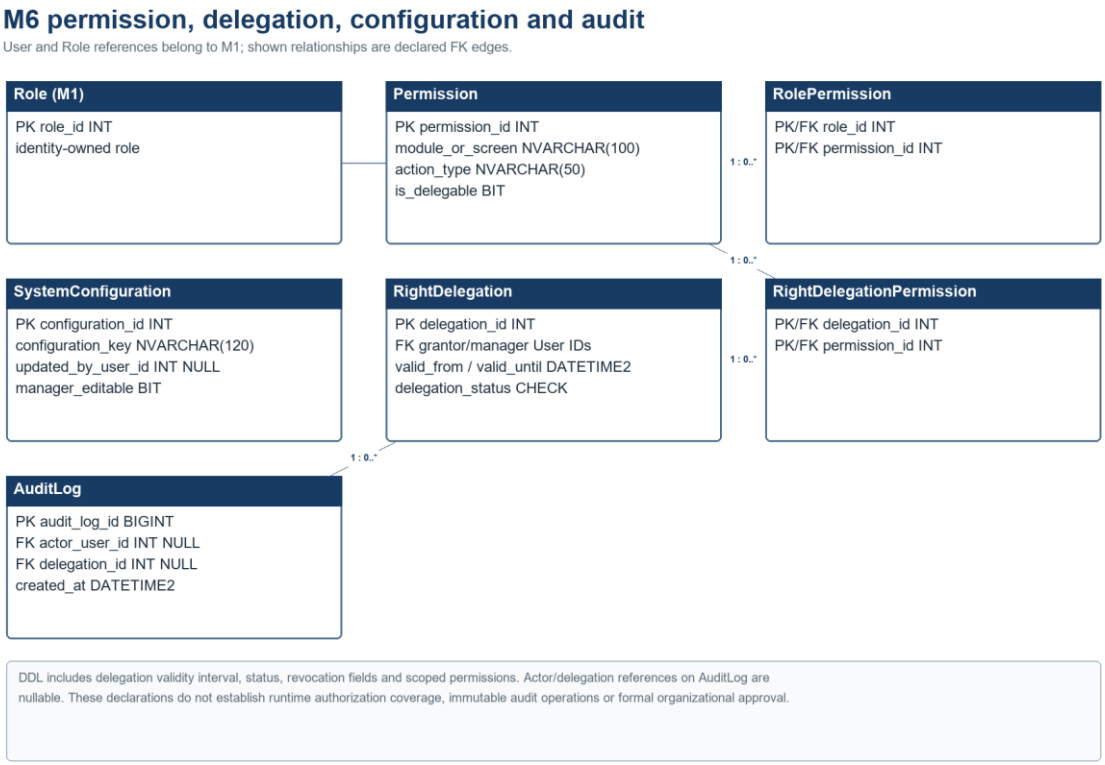


Figure 10. Administration and permission ERD. Role and User are M1 references; delegation scope and nullable audit references follow the DDL.

Selected columns and FK edges only; consult the existing static database dictionary for the full schema inventory. This figure describes source DDL, not live deployment.

Selected table descriptions and key constraints

Module	Representative tables	Important keys / constraints	Boundary / verification status
M1 Identity	User, AuthIdentity, UserRole, CustomerProfile, HealthProfile, ConsentRecord, HealthDataDeletionRequest	UserRole composite PK; AuthIdentity provider/subject unique; consent/deletion checks.	Health data and consent remain subject to privacy decisions; no production state claimed.
M2 Booking	RetreatPackage, ResortZone, VillaCategory, PhysicalVilla, RetreatBooking, ResidenceRecord	Booking FKs to package/customer/category; assigned villa nullable; villa category distinct from physical villa.	No schema FK from PhysicalVilla.zone_name to ResortZone.
M3 Spa	SpaSession, TherapistTreatmentQualification, TreatmentRoomCompatibility, SpaScheduleHistory, SpaDispatchOutbox	Resource FKs; composite qualification/compatibility PKs; guarded history/outbox declarations.	SQL Server locking and scheduler/provider runtime unverified.
M4 F&B	MenuItem, MenuItemIngredient, IngredientBatch, MealOrder, MealOrderItem, PackageFnBEntitlement	Recipe join composite PK; expiry/quantity checks; order type/status checks; one entitlement per package.	Inventory and entitlement deployed behavior unverified.
M5 Billing	Folio, FolioItem, Invoice, PaymentTransaction, AdjustmentRequest, CreditNote	Unique invoice-per-folio; nullable source/payment references; filtered unique source indexes; financial checks.	SQL Server index enforcement, gateway, payment and checkout runtime unverified.
M6 Admin	Permission, RolePermission, SystemConfiguration, RightDelegation, RightDelegationPermission, AuditLog	Composite role/delegation permission keys; bounded validity fields; nullable audit references.	DDL does not establish runtime enforcement, audit immutability, or formal approval.

The static baseline contains 52 CREATE TABLE declarations (50 unconditional and two guarded). Entity annotations and DDL are source evidence only and do not prove deployed SQL Server state.

II. Detailed Code Design

The following module descriptions synthesize Phase 0.1-F registers A01–H02, existing SDS V2 registers, and targeted static inspection of current working-tree source. The repository HEAD recorded in Document Control is fcd628eb32679f75fef385f1f656a07c83f1a5aa; the inspected working tree already contains pre-existing source changes. Those changes were read only and were not executed. Component and method statements are static observations; absent trace joins are not inferred.

1. M1 Authentication and Sensitive Health Profile

M1 handles identity/account flows, customer profile, consent and health. AuthController/AuthService, HealthProfileController/HealthProfileService, ConsentController/ConsentService and User/AuthIdentity/CustomerProfile/HealthProfile/ConsentRecord are source candidates.

Responsibilities and design constraints

- JWT filter/security chain and selected method guards exist. Backend authorization is authoritative; UI guards do not secure APIs. Full route-level 401/403 evidence is incomplete.
- Apply consent, retention and minimum-access principles. Therapist sees treatment-relevant health/safety context; F&B staff sees relevant allergy/dietary information; Receptionist must not have unrestricted health access.
- Health deletion may be deferred while an active/upcoming stay needs the data. Revision 1.1 does not specify retention periods or detailed consent semantics; do not invent them.
- Fail closed where ownership, consent or safety-critical data cannot be established. Exact error mapping and rollback evidence are endpoint-specific and incomplete.

Evidence status

PARTIAL — source exists; privacy retention, field-level approval, full trace and runtime access remain unverified.

Components and operations

AuthController/AuthService implement authentication entry points; HealthProfileController/HealthProfileService handle profile operations; ConsentController/ConsentService handle consent records. User, AuthIdentity, CustomerProfile, HealthProfile, and ConsentRecord remain separate source concepts. Shared SecurityConfig/JWT infrastructure authenticates requests; selected controllers/services consume the principal. This is not a complete route-to-data chain.

Interfaces, data access, and failure boundary

Identity/profile repositories are the persistence path in the source inventory. M3/M4 health projections must remain purpose-limited. Consent, retention, deletion scheduling, field-level authorization, and endpoint-by-endpoint 401/403/error behavior lack complete evidence. Fail closed when ownership, consent, or safety context is unknown; no retention period is specified.

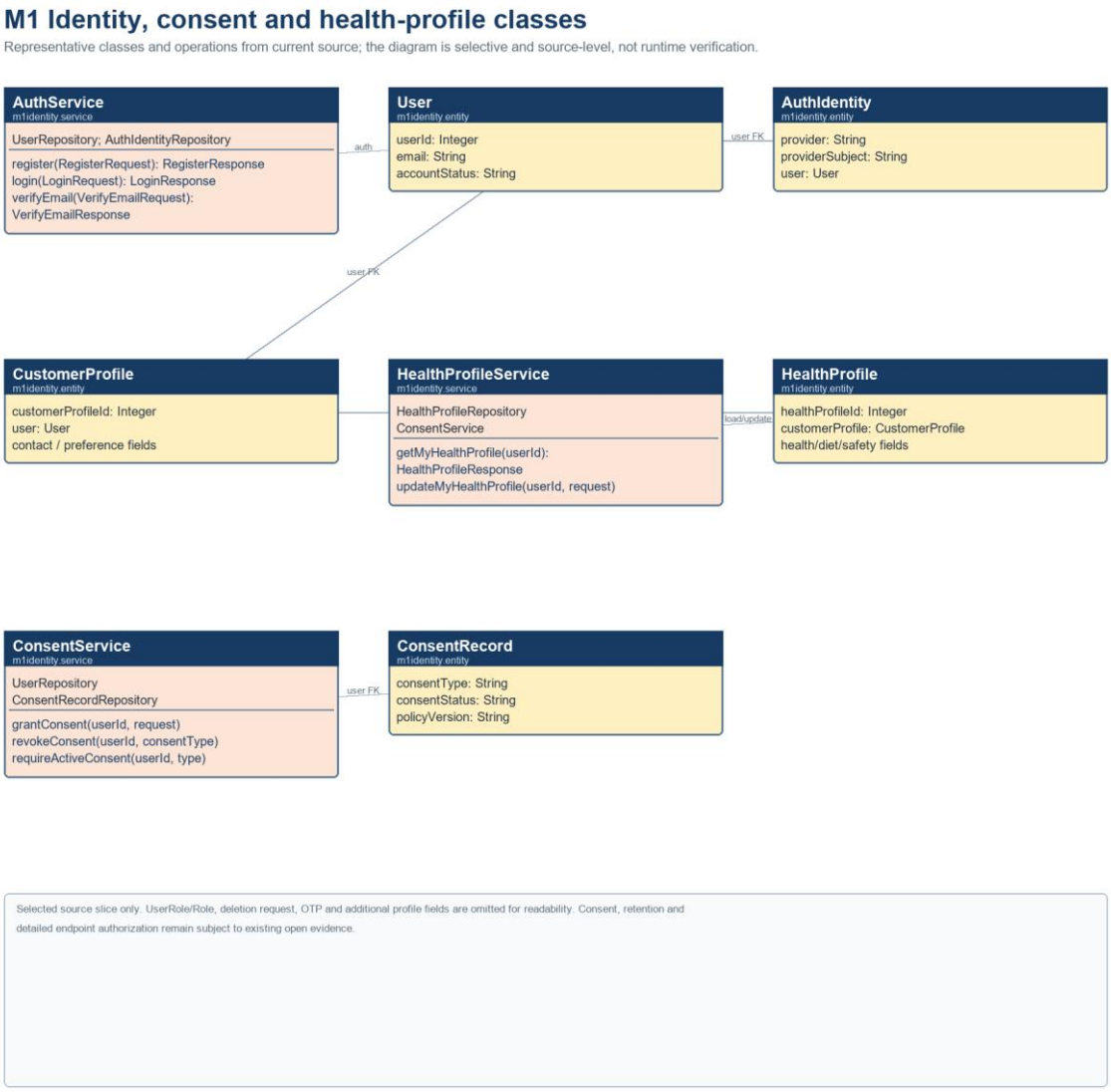


Figure 11. M1 identity, consent and health-profile classes. Selected fields and operations are source-level; privacy policy and deployed authorization remain unverified.

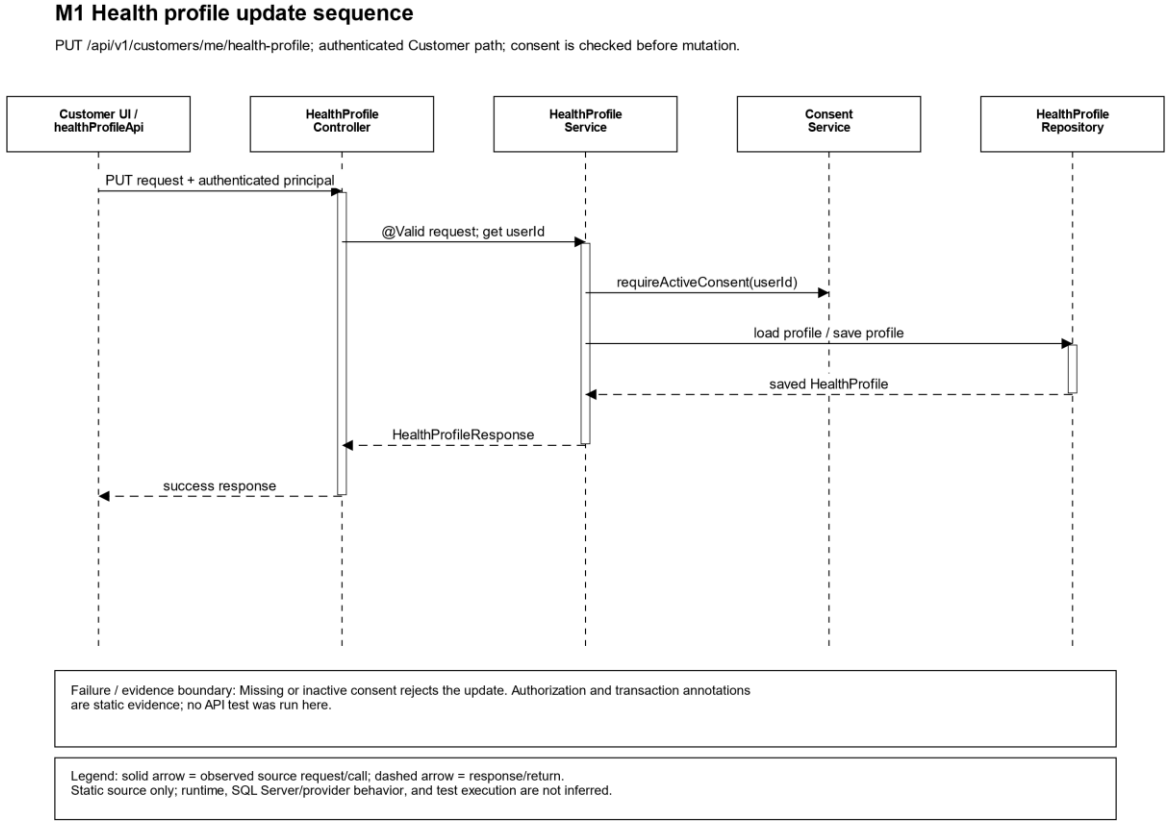


Figure 12. Customer health-profile read/update path; authorization and consent behavior are source-level evidence only. Selected path only; error and alternate branches are not exhaustive.

2. M2 Retreat Package and Accommodation Booking

M2 manages retreat packages, booking, villa inventory and reception workflows. RetreatBookingController/Service, VillaInventoryController and ReceptionistOperationController/Service are source-listed. A Villa Category is distinct from a Physical Villa.

Responsibilities and design constraints

- Revision 1.1 assigns Booking and deposit business rules to M2, while PaymentTransaction ownership is M5. Existing path compliance is not proven.
- Reception check-in source checks booking/villa conditions and records residence state; locking methods exist. SQL Server concurrency is unverified.
- ResortZone, VillaCategory and PhysicalVilla occur in source/schema. Do not infer zone allocation or package entitlements from entity shape.
- Role/actor checks occur in selected code; complete endpoint permissions, ownership rules and error responses need trace evidence.

Evidence status

PARTIAL — code paths exist; M2 rules, schema relationship, payment integration and SQL Server behavior remain incomplete.

Components and operations

RetreatBookingController/Service own booking-facing paths; ReceptionistOperationController delegates check-in and walk-in work to ReceptionistOperationService; VillaInventoryController

exposes villa state. assignVillaAndCheckIn checks booking, physical-villa, and residence conditions, persists stay state, and records audit in the reviewed service path. VNPay deposit code is adjacent but does not settle OI-04.

Interfaces, data access, and failure boundary

The check-in controller maps AssignVillaRequest to BookingSummaryResponse; transaction and repository lock methods are visible in selected service paths. VillaCategory is distinct from PhysicalVilla and ResortZone is separately represented; names do not define allocation rules. SQL Server concurrency is unverified. Role and ownership review remains incomplete across routes.

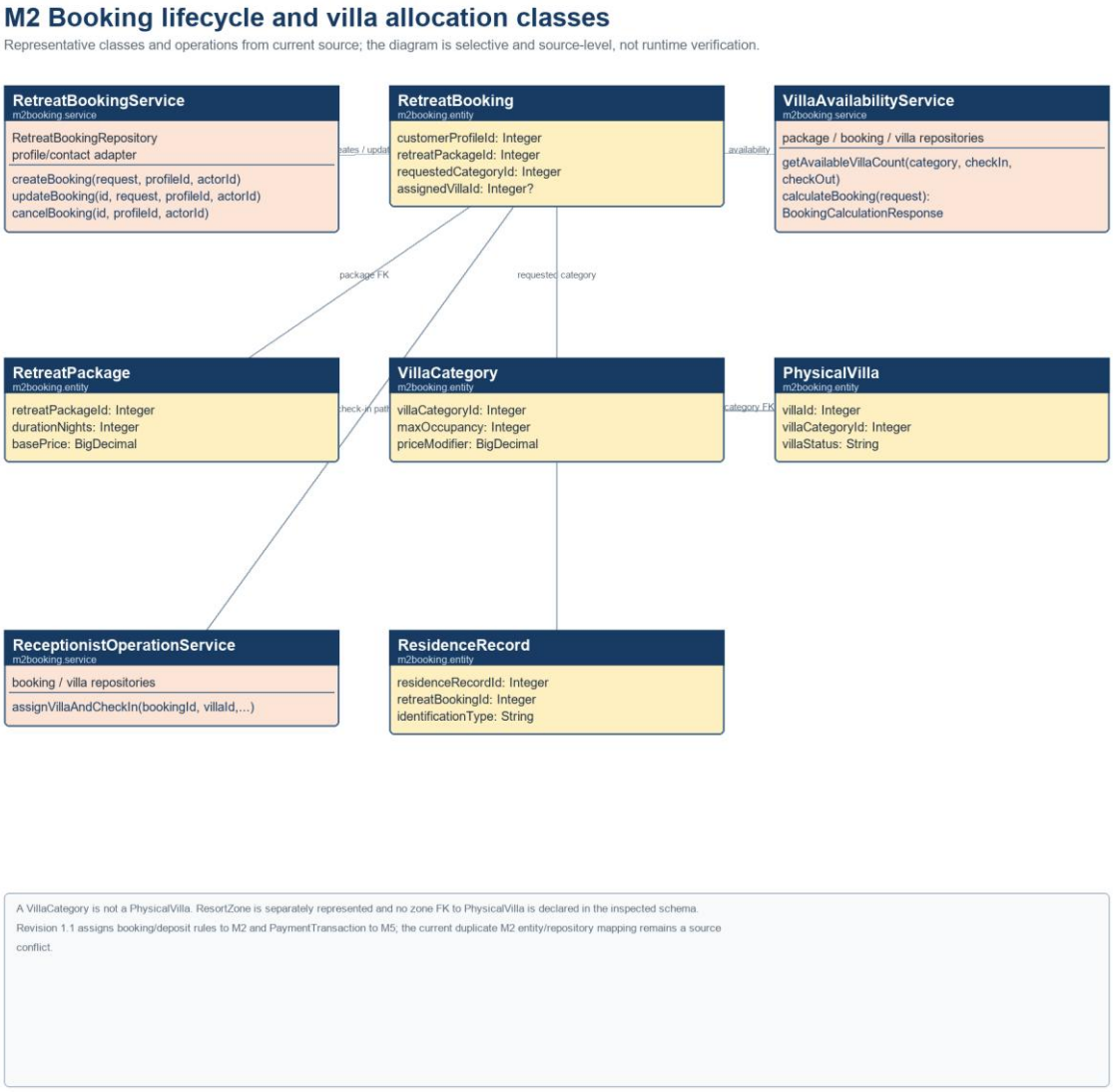


Figure 13. M2 booking lifecycle and villa allocation classes. Source-level relationships only; SQL Server concurrency and the M2/M5 PaymentTransaction ownership conflict remain open.

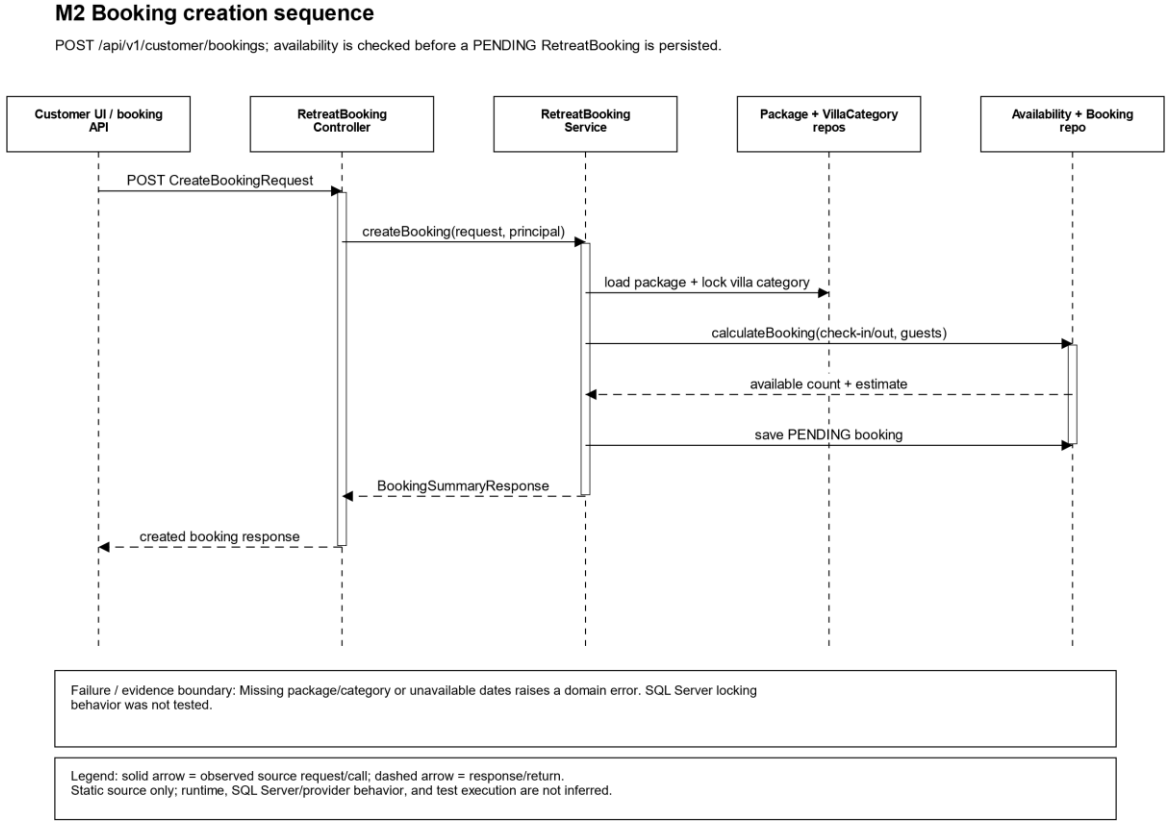


Figure 14. Booking creation path through availability calculation and persistence; SQL Server concurrency is unverified. Selected path only; error and alternate branches are not exhaustive.

3. M3 Spa and Therapy Scheduling

M3 scheduling treats candidate resource as an indivisible bundle: time slot, qualified Therapist/Instructor and compatible TreatmentRoom. Candidate generation is read-only; final reservation revalidates and reserves atomically. Components include SpaBookingController, SchedulingEngineService, SpaSessionService, SessionRecoveryService, SessionTimelineScheduler and SpaDispatchOutboxWorker.

Responsibilities and design constraints

- Resource conflict must not create partial reservation. Existing Customer session remains valid while browsing reschedule candidates; replacement is effective only after successful transactional replacement.
- Persisted statuses: SCHEDULED, RESCHEDULING, CANCELLED, COMPLETED, NO_SHOW. Separate confirmation states: PENDING, CONFIRMED, DECLINED, EXPIRED. Persisted IN_PROGRESS remains prohibited.
- Cross-customer recovery/swap requires explicit mutual consent (Rev 1.1). Existing mutual-consent workflow is PARTIAL; do not claim implemented.
- OI-02 Option B: five attempts total, delays 30s/1m/2m/4m. Ambiguous SMTP outcome must be reconciled, not blindly resent. Historical ADR reconciliation remains necessary.
- Calendar adapter lacks authenticated API client. M3→M5 adapter uses non-idempotent postFolioCharge and catches/logs exceptions; no durable retry is evidenced. M5 idempotent target contract is not yet enforced by this path.

Evidence status

PARTIAL — H2 evidence only; SQL Server locks, live SMTP/Calendar, OI-01 remediation and cross-customer consent remain unverified.

Components and operations

SpaBookingController.generateCandidates calls read-only SchedulingEngineService.generateCandidates; confirmReservation calls SpaSessionService.confirmReservation. SpaSessionService owns attendance confirmation, decline, cancellation, and reschedule paths. SessionRecoveryService and schedulers evaluate disrupted or due sessions; SpaDispatchOutboxWorker claims persisted work and delegates to email/calendar adapters. CustomerStayPort and health projection adapters read M2/M1 context; the billing adapter calls M5 directly.

Interfaces, data access, and failure boundary

GenerateCandidatesRequest requires treatmentId and preferredDate; ConfirmReservationRequest requires treatmentId, therapistId, roomId, start, and end and validates interval shape. Candidate generation does not reserve. Final reservation is transactional in source and revalidates/locks resources; SQL Server behavior remains unverified. Confirmation and session statuses are separate; IN_PROGRESS is not persisted. Current worker defaults are configurable, but scheduler, claims, SMTP outcome certainty, and Calendar delivery were not run. SPA billing still calls non-idempotent postFolioCharge, catches/logs failures, and shows no adapter-level durable retry (OI-01).

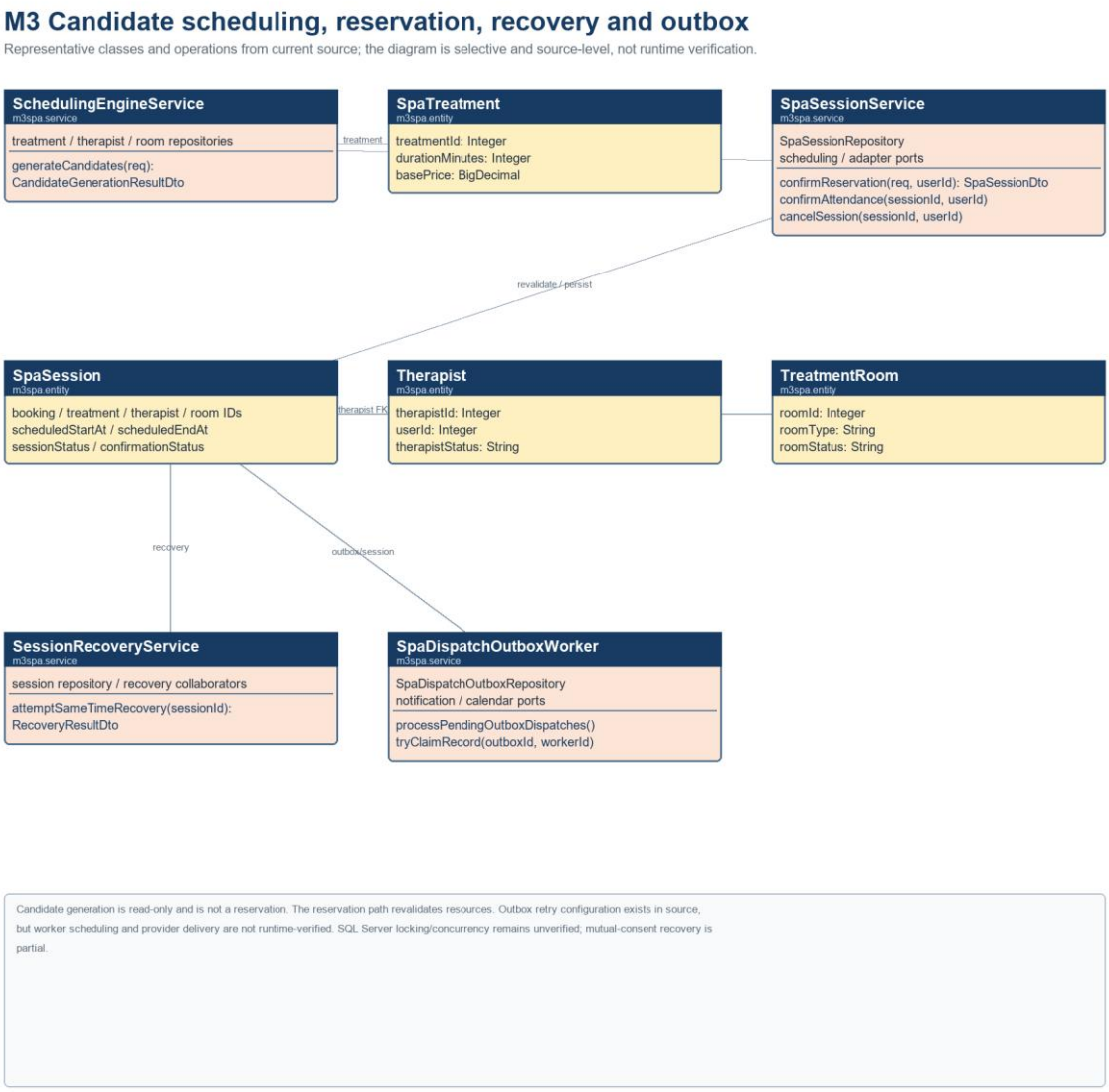


Figure 15. M3 candidate scheduling, reservation, recovery and outbox classes. Candidate generation is read-only; the reservation path revalidates resources. Runtime and SQL Server behavior remain unverified.

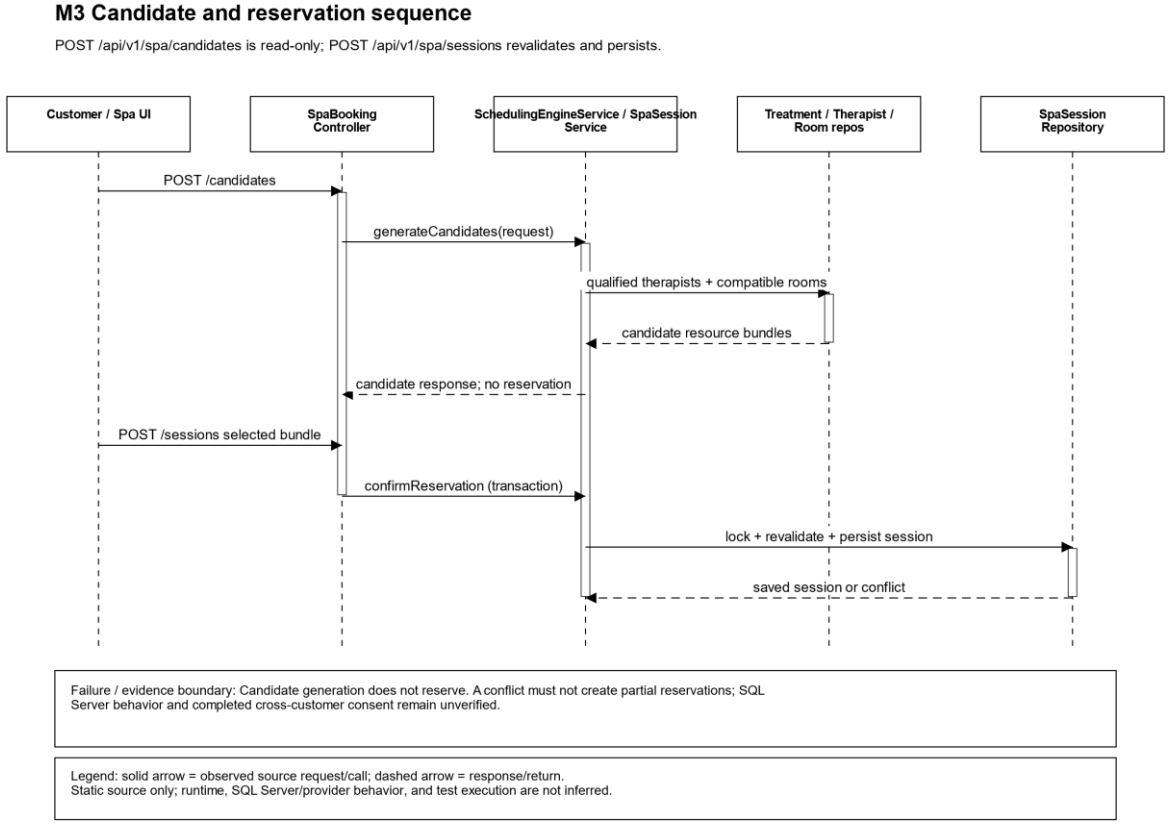


Figure 16. Read-only candidate generation followed by transactional reservation revalidation; SQL Server behavior is unverified. Selected path only; error and alternate branches are not exhaustive.

4. M4 Dietary F&B Management

M4 manages menu/order, KDS, recipe and ingredient inventory. Source components include MenuController, FnbKdsController, MenuService and FnbKdsOrderServiceImpl; entities include MealOrder/Item, MenuItem and IngredientBatch.

Responsibilities and design constraints

- Allergy suitability must fail closed. If suitability cannot be established, do not treat a meal as suitable; do not reduce this to a warning.
- Inventory must respect usable quantity, expiry and condition; reserve/deduct at the approved preparation transition, not an invented earlier state.
- G04 lists PENDING → PREPARING → READY → DELIVERED and CANCELLED. Full transition and authorization matrix is partial.
- Eligible A_LA_CARTE delivery calls M5 postFolioChargeIdempotent with sourceMealOrderId. Both service methods carry transactional annotations; runtime propagation and SQL Server concurrency were not tested.
- OI-06 uses explicit configurable service-location contract; OI-07 uses package-specific entitlement matrix with effective dates. These are target decisions; implementation/configuration evidence must be mapped.

Evidence status

PARTIAL — source and test files exist; location, entitlement, inventory and allergy runtime evidence remain incomplete.

Components and operations

MealOrderController exposes customer ordering and staff status updates. FnbKdsOrderServiceImpl places orders, checks booking context, reads the kitchen queue, locks an order during status change, rejects changes after DELIVERED/CANCELLED, and handles delivery/cancellation. DefaultBookingAndDietaryIntegrationService reads M2 booking and M1 customer/health data. IngredientBatch and menu/recipe entities support inventory; M6 audit calls are present in selected paths.

Interfaces, data access, and failure boundary

MealOrderController routes under /api/v1/fnb/orders call MealOrderService; its status endpoint uses UpdateOrderStatusRequest and returns KitchenOrderResponse. Source traces the FnbKdsController status route through MealOrderService.updateOrderStatus to FnbKdsOrderServiceImpl; eligible A_LA_CARTE delivery invokes M5 postFolioChargeIdempotent with sourceMealOrderId, which is passed into FolioItemEntity and queried by the duplicate-check repository. Both service methods are transactional in source, but runtime propagation and SQL Server races remain unverified. Allergy suitability must fail closed. The full actor/transition matrix, cross-bean propagation, and SQL Server races remain unverified; service-location and package-entitlement matrices remain target decisions.

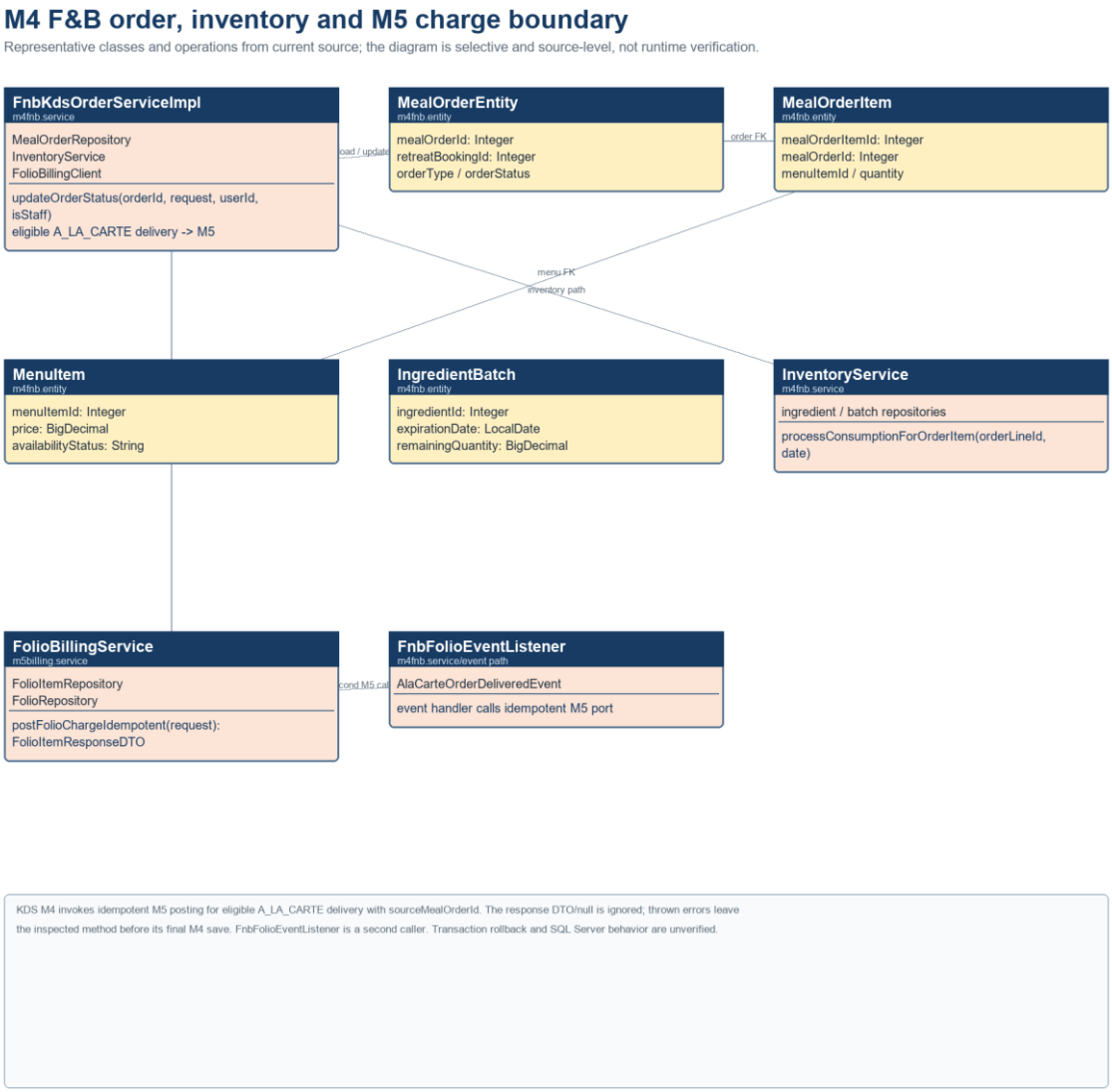


Figure 17. M4 order, inventory and M5 charge classes. The direct KDS caller and second event-listener caller are both shown; idempotency and transaction behavior are not runtime-verified.

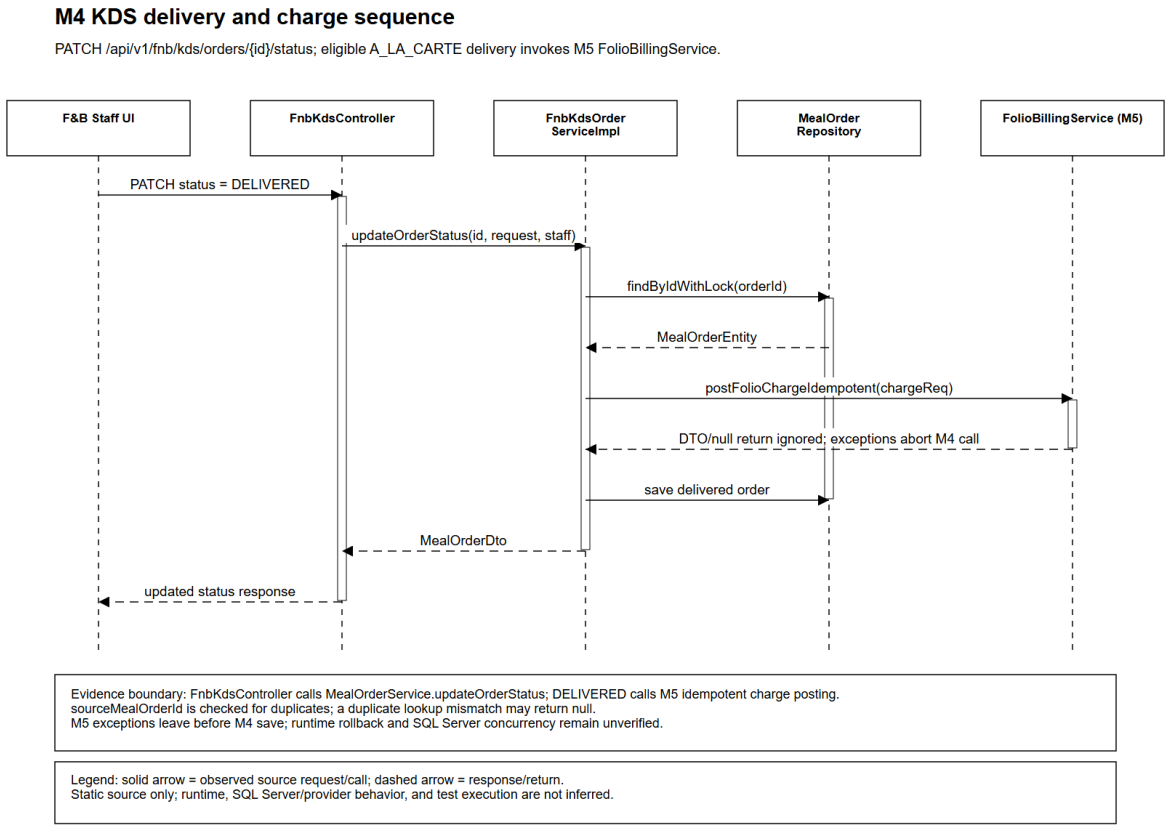


Figure 18. KDS delivery calls M5 for eligible A_LA_CARTE charges. Returned DTO or null is ignored; a thrown M5 exception exits before M4 status save. Runtime rollback and SQL Server behavior remain unverified. Selected path only; error and alternate branches are not exhaustive.

5. M5 Consolidated Checkout and Analytics

M5 contains Folio, Invoice, PaymentTransaction and checkout services. Revision 1.1 selects M5 ownership of PaymentTransaction and an M5-owned idempotent Folio posting contract. These target ownership directions do not prove every present code path conforms.

Responsibilities and design constraints

- Folio is a mutable running stay account; Invoice is a formal snapshot. A successful PaymentTransaction is not by itself proof of checkout completion: an asynchronous VNPay callback closes the stay only when the invoice becomes PAID and the billing guard passes; finalizeCheckout separately checks balance and guards.
- Checkout must block when unpaid/unresolved Spa or F&B charges remain. CheckoutService source locks/rechecks booking/folio and checks balance, but SQL Server locking and charge visibility are unverified.
- M3 adapter calls a non-idempotent M5 operation and catches/logs errors. M4 eligible delivered A_LA_CARTE uses an idempotent source order key. OI-01 risk remains until source changes and verification demonstrate M5 target contract across paths.
- VNPay is the only in-scope payment method; Cash/POS and other methods are deferred. No live VNPay verification. Refund, adjustment, settlement and reconciliation details need evidence; no policy is invented.

Evidence status

PARTIAL — target ownership selected; adapter gap, gateway runtime, refunds and SQL Server behavior remain.

Components and operations

FolioBillingService posts charges and creates invoice snapshots; CheckoutService.processPayment evaluates payment state, and finalizeCheckout validates outstanding amounts before stay close. The observed finalization path reads invoice details, locks Folio and RetreatBooking via repositories, sums successful PaymentTransaction values and final credit notes, then applies close state. VNPayService.createPaymentUrl persists a PENDING PaymentTransaction; verify-return processes the callback through confirmGatewayPayment. A successful callback can close the Folio and Booking when the Invoice becomes PAID and the billing guard passes. Runtime gateway behavior is unverified.

Interfaces, data access, and failure boundary

POST /api/v1/billing/charges calls non-idempotent postFolioCharge; postFolioChargeIdempotent is a separate source-key method used by M4 delivery. This does not prove the M3 path meets OI-01. PaymentRequestDTO source accepts a broader method vocabulary than the VNPay-only strategic scope; source validation does not prove deferred methods are operational or approved. Folio is mutable; Invoice is a snapshot. A successful PaymentTransaction is distinct from stay closure: the callback closes only on PAID invoice plus passing billing guard; finalize-checkout separately rejects remaining balance or billing-guard failures. SQL Server concurrency, callbacks, refunds, and reconciliation remain unverified.

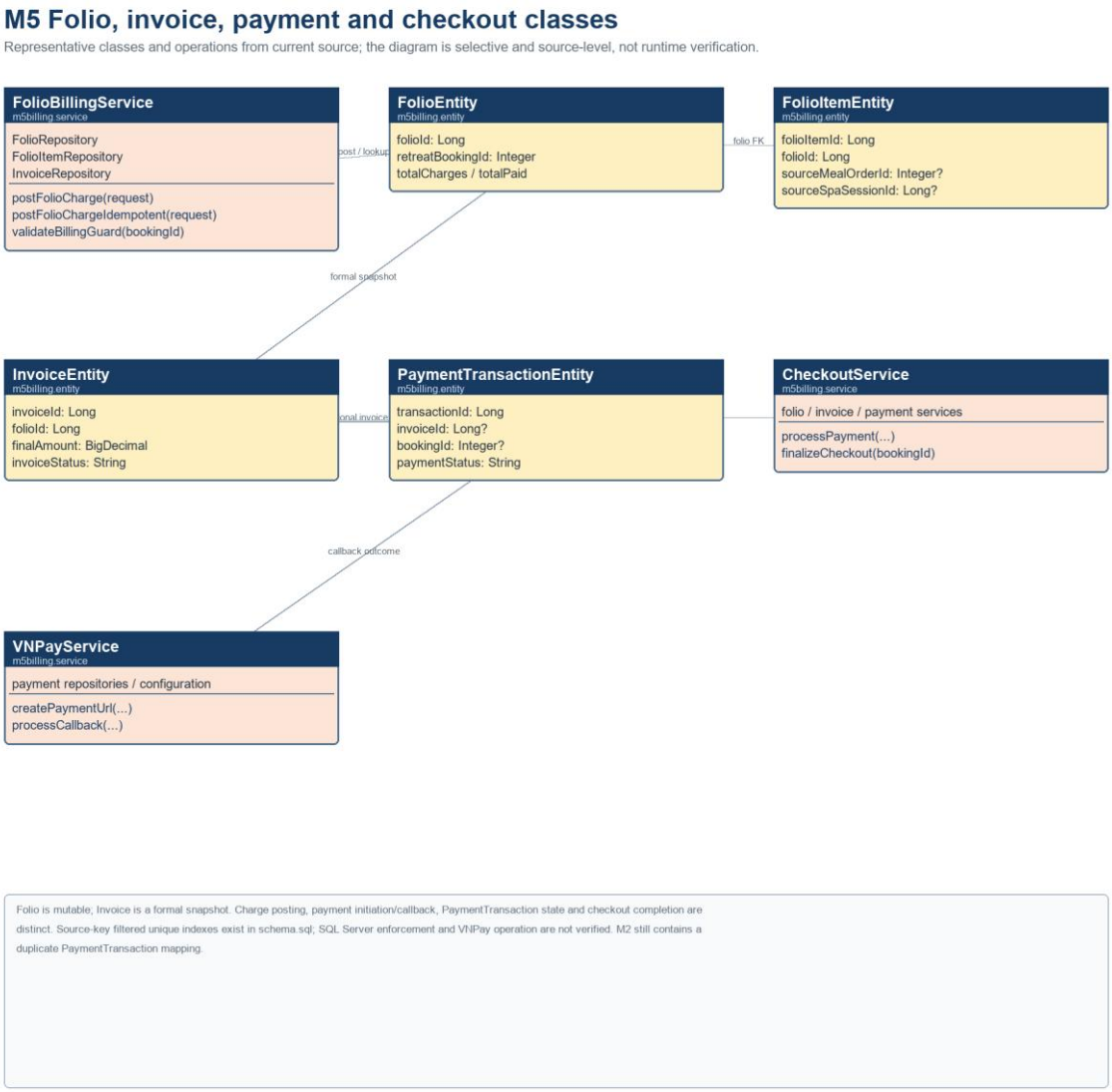


Figure 19. M5 folio, invoice, payment and checkout classes. Charge, payment, callback and checkout are separate source paths; SQL Server and VNPay behavior remain unverified.

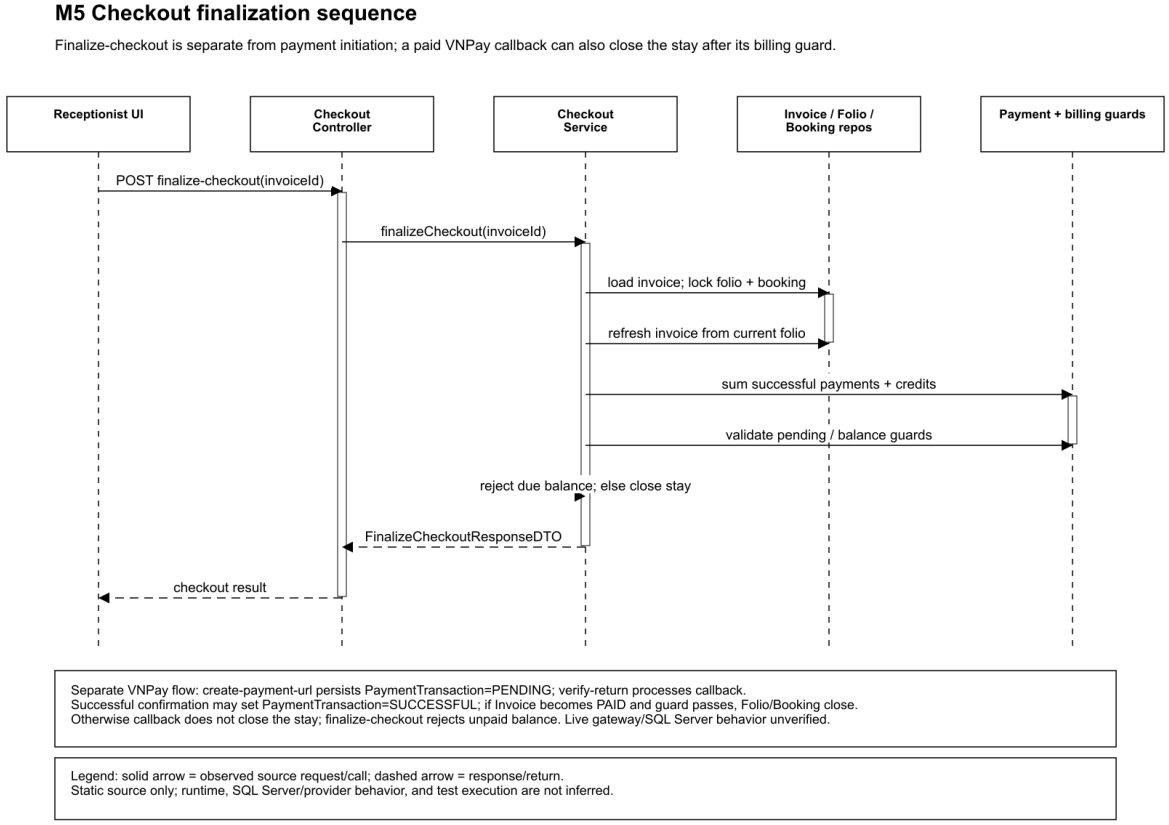


Figure 20. Invoice-owned finalization path plus separate VNPay callback outcome; a paid, guarded callback may close the stay. Live gateway and SQL Server behavior remain unverified. Selected path only; error and alternate branches are not exhaustive.

6. M6 System Administration and Permission Management

M6 includes permission catalog and role mapping, permission evaluation, staff, delegation, audit and system configuration. Static components include AdminPermissionController, AdminDelegationController, PermissionEvaluator and DelegationService.

Responsibilities and design constraints

- Backend authorization is authoritative; selected endpoints use permission guards, but full route coverage is incomplete.
- Delegation follows least privilege, is time-limited and auditable. Broad, indefinite or unaudited delegation is not an acceptable target.
- Canonical SystemConfiguration and RightDelegation API/data/UI contract is not established by the available evidence. Do not assume frontend shape is the canonical contract.
- Audit writes are present in selected paths. Retention, immutability, read access and operation are not runtime verified.

Evidence status

PARTIAL — source exists; contract, all-consumer coverage, policy values and formal security approval remain pending.

Components and operations

SecurityConfig enables method security; PermissionEvaluator delegates to PermissionService. AdminDelegationController applies S52 view/create/delete checks to list/create/revoke. DelegationService validates actor, Manager target, permission and time bounds, and records audit in selected create/revoke paths. SystemConfiguration and RightDelegation repositories back configuration/delegation areas, but canonical API/data/UI contracts are not established.

DelegationCreateRequest uses @Valid; create returns DelegationView and revoke returns updated DelegationView. Status refresh/revoke source exists, but scheduler invocation and all module consumers are not verified. Permission evaluation is intended to deny on failure; selected guards do not prove complete route coverage. Audit retention/immutability and policy values require security and organizational approval. Frontend guards do not replace backend authorization.

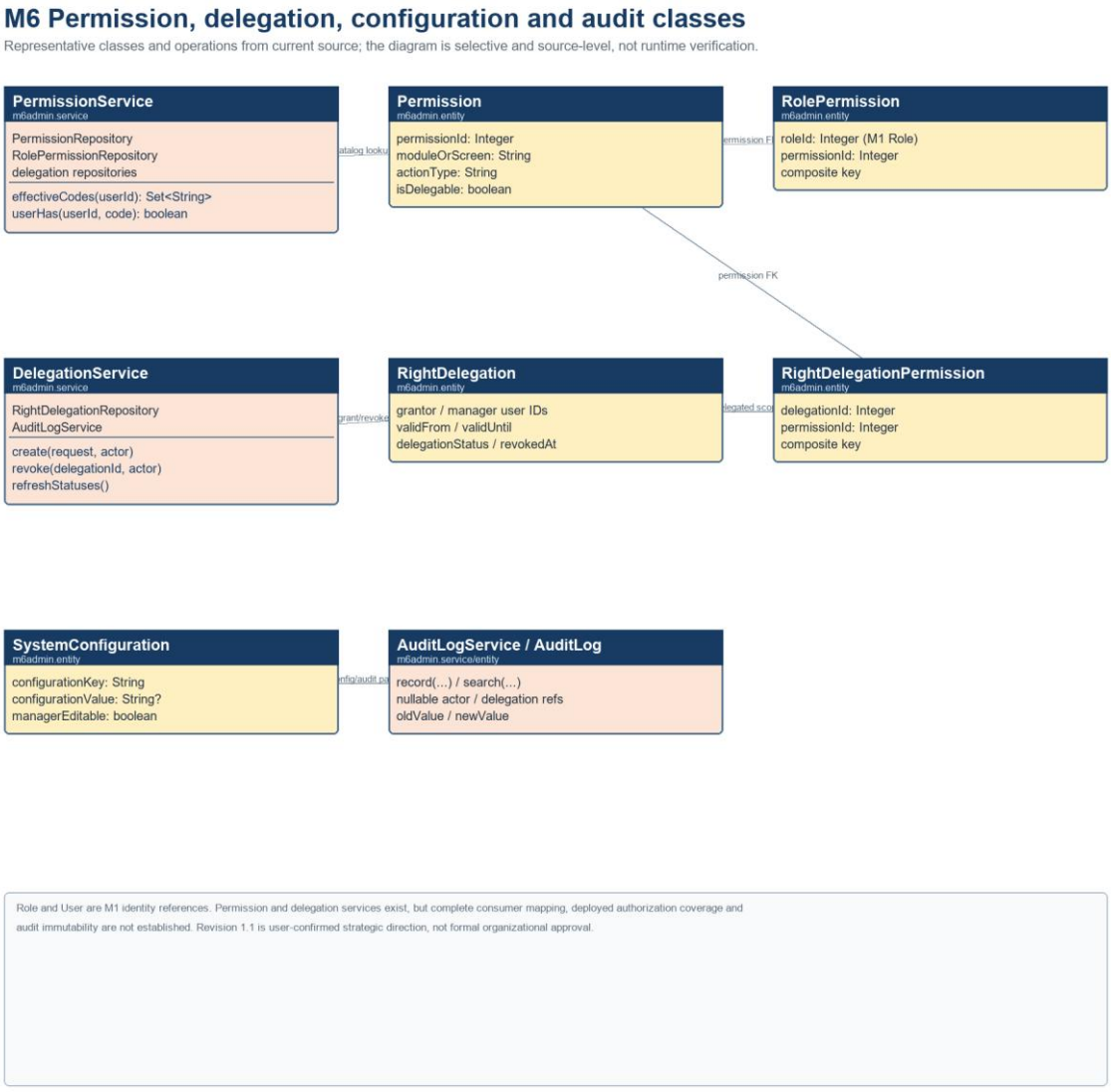


Figure 21. M6 permission, delegation, configuration and audit classes. Source structure only; complete consumer mapping, runtime authorization coverage and audit guarantees remain unverified.

Interfaces, data access, and failure boundary

AdminPermissionController/AdminDelegationController expose selected permission and delegation routes with @perm guards. M6 uses its permission, delegation and configuration repositories and reads M1 UserRepository/RoleRepository data; M1 calls M6 PermissionService.effectiveCodes and

selected parameter/audit services. Delegation validation/access errors and selected audit writes are source-visible. Full route coverage, scheduler/runtime behavior and audit guarantees remain unverified.

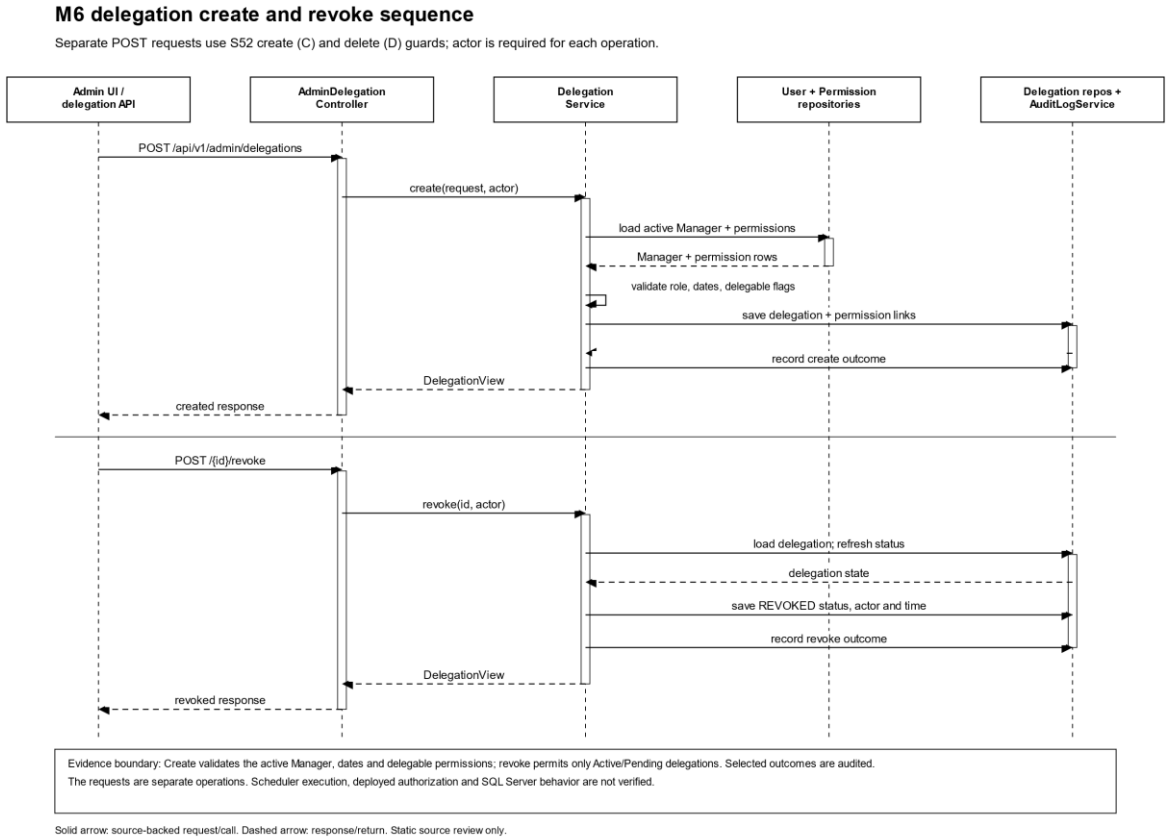


Figure 22. Delegation creation/revocation path; scheduler and deployed authorization are not runtime-verified. Selected path only; error and alternate branches are not exhaustive.

III. Other Design Specifications

Document status and evidence scope: Version 3.0. Verified implementation facts are limited to inspected repository source and static artifacts. Revision 1.1 records user-approved strategic directions, not formal organizational approvals. Proposed contracts, unverified runtime/deployment behavior, and unresolved gaps remain explicitly labelled. No tests, application, database, SQL Server, SMTP, Google Calendar, or VNPay were executed for this specification.

1. Security and privacy

- JWT authentication and backend role/permission checks are source-backed. Full effective-route coverage and deployed 401/403 outcomes are not established. Frontend guards provide presentation only.
- Apply least privilege, resource ownership checks for Customer/Therapist records, and minimum health-data projection. Health access remains purpose-scoped; exact field matrices need approved evidence.
- Consent, identity, profile and health are distinct concerns. Deletion can be deferred when active/upcoming stay needs data; no retention period or grace window is invented.
- Never include credentials or machine-specific values in specifications. Secret storage and operational identity controls require an approved deployment design.

2. Interfaces, errors, and transaction boundaries

REST controllers, module services, Spring Data repositories, selected ports/adapters and scheduled workers are visible in source. The 152 route candidates match handler/path annotations, but do not form a complete effective API contract. Authentication, composed routes, DTO validation, error mapping, UI callers, repository writes and tests are not joined for every route.

Boundary	Observed path	Failure/design treatment	Status or target
M2→M3	DefaultCustomerStayAdapter reads M2 booking and M1 customer profile via read-only transaction/JDBC fallback.	Returns empty when context unavailable (fail closed); active bean and caller response unverified.	Static evidence; runtime pending.
M3→M5	DefaultBillingAdapter calls non-idempotent postFolioCharge with session source; catches/logs.	Durable retry/idempotency not shown; reconciliation risk.	Target: M5-owned idempotent contract (OI-01); implementation gap.
M4→M5	Eligible A_LA_CARTE calls postFolioChargeIdempotent with sourceMealOrderId.	Idempotent method/source check exists; transaction and SQL Server race untested.	Target owner M5; accounting ownership path still must be reconciled.
M1→M4	Dietary integration reads health profile.	Fail closed where allergy suitability cannot be established; minimize projection.	Field-level runtime authorization pending.
M6→modules	Permission evaluator and selected @perm guards.	Deny on evaluation failure; all route consumers not mapped.	Least privilege target; coverage pending.

A proposed API error taxonomy is validation 400, authentication 401, authorization/ownership 403 (or non-disclosing not-found where approved), not-found 404, lifecycle/conflict 409, and dependency failure 5xx or safe domain error. This is a proposed consistency contract; exact endpoint behavior must be verified from controller advice and API tests. The selected interface table is a high-risk subset: it records declarations and visible calls, not 152 end-to-end contracts. Response, validation, exception, and security fields are marked unknown when not confirmed in inspected source.

Declared endpoint and handler	Request → response types	Visible access / validation	Service path and evidence limit
GET /api/v1/customers/me/health-profile HealthProfileController.getMyHealthProfile	No request body → ApiResponse<HealthProfileResponse>	@PreAuthorize hasRole(CUSTOMER); identity from authenticated principal.	HealthProfileService.getMyHealthProfile; field-level access, error mapping, and runtime outcomes not verified.
POST /api/v1/spa/candidates SpaBookingController.generateCandidates	GenerateCandidatesRequest → CandidateGenerationResultDto	SPA-role matcher in SecurityConfig; @Valid. No method @PreAuthorize visible.	SchedulingEngineService.generateCandidates; read-only. DTO validation failure responses and runtime contract not verified.
POST /api/v1/spa/sessions SpaBookingController.confirmReservation	ConfirmReservationRequest → SpaSessionDto	SPA-role matcher; @Valid; principal passed to service. No method guard visible.	SpaSessionService.confirmReservation; transactional source path; rollback, error mapping, and SQL Server behavior not verified.
POST /api/v1/receptionist/check-in/{bookingId}/assign-villa assignVillaAndCheckIn	AssignVillaRequest → BookingSummaryResponse	Receptionist route matcher; no controller method guard visible in inspected handler.	ReceptionistOperationService.assignVillaAndCheckIn; lock methods in source; error mapping and SQL Server behavior not verified.
PATCH /api/v1/fnb/orders/{orderId}/status MealOrderController.updateOrderStatus	UpdateOrderStatusRequest → KitchenOrderResponse	@PreAuthorize FNB_STAFF / F&B STAFF; @Valid; status required.	MealOrderService.updateStatus. Link to FnbKdsOrderServiceImpl Folio posting remains unverified; transition errors/transaction behavior need reconciliation.
POST /api/v1/billing/charges FolioBillingController.postFolioCharge	PostChargeRequestDTO → FolioItemResponseDTO	RECEPTIONIST/ADMIN @PreAuthorize; @Valid.	FolioBillingService.postFolioCharge; non-idempotent; OI-01 risk. Error mapping and durable retry not evidenced.
POST /api/v1/billing/invoices/{invoiceId}/pay CheckoutController.processPayment	PaymentRequestDTO → PaymentResponseDTO	Staff roles or invoice-access evaluator; @Valid.	CheckoutService.processPayment; transaction in source; payment errors/callbacks and live gateway unverified.
POST /api/v1/billing/invoices/{invoiceId}/finalize-checkout CheckoutController.finalizeCheckout	No body → FinalizeCheckoutResponseDTO	@PreAuthorize billingAccess.canFinalizeCheckout; effective outcomes not run.	CheckoutService.finalizeCheckout; locks Folio/Booking and checks amounts; response errors and SQL Server isolation unverified.
POST /api/v1/admin/delegations AdminDelegationController.create	DelegationCreateRequest → DelegationView	@perm.has(S52,C); @Valid; CurrentActor principal.	DelegationService.create; validation/audit static; error mapping, all consumers, and policy approval not verified.

Figure 3 arrows point from caller to dependency: M3 reads M1/M2 context and calls M5; M4 reads M1/M2 context and calls M5/M6. M1 AuthController calls M6 PermissionService.effectiveCodes; M1 AccountLockoutService calls M6 BusinessParameterService and AuditLogService. M6 PermissionEvaluator and DelegationService read M1 UserRepository/RoleRepository data. No direct M3-to-M4 call was found in the reviewed scan.

3. Lifecycle and concurrency

Figure 16 — M3 reservation, confirmation, recovery, and dispatch

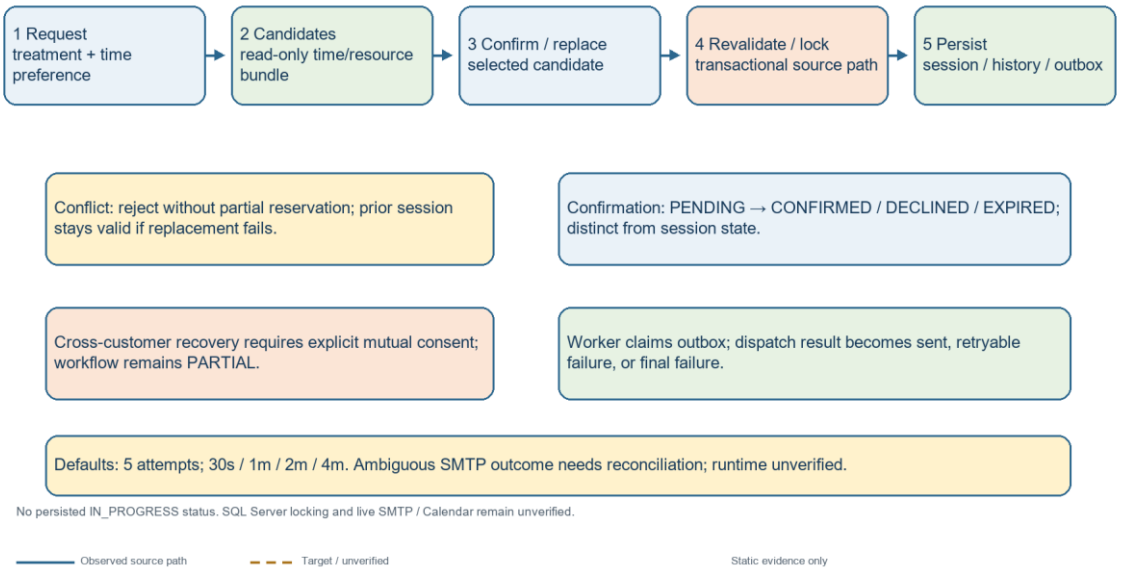


Figure 23. M3 reservation, confirmation, recovery, and dispatch. SQL Server and provider behavior remain unverified.

Figure 17 — M5 charge sources, payment, and checkout guard

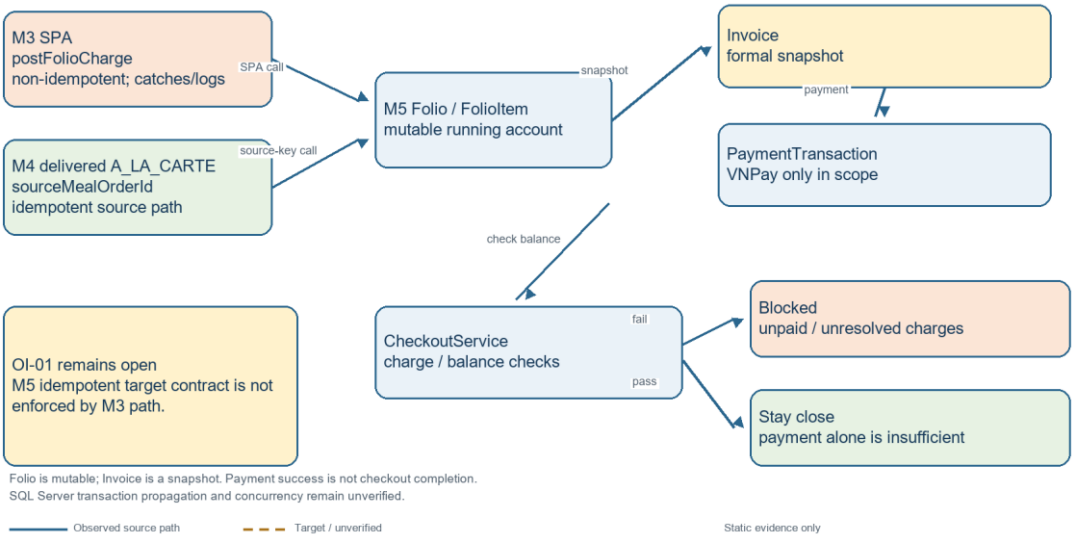


Figure 24. M5 charge and checkout paths. M3 and M4 adapter semantics differ; payment success is not checkout completion.

M3 search does not reserve; confirmation revalidates and locks the resource bundle. Keep original session while a replacement is pending and commit replacement atomically. Cross-customer recovery requires both parties' explicit consent. Checkout must not omit delivered F&B charges. H2 cannot establish SQL Server locking, isolation, deadlocks, triggers or production scheduler timing.

Figure 23 expands the reservation and dispatch flow. Current working-tree defaults specify five total attempts and delays of 30 seconds, 1, 2, and 4 minutes. This is static configuration evidence, not

worker-activation or delivery proof. Ambiguous SMTP outcomes require reconciliation. The SPA billing adapter still calls non-idempotent postFolioCharge and catches/logs failures.

4. External integrations and background processing

Integration / trigger	Direction and source behavior	Failure / retry evidence	Verification status
M3 Email / SMTP	M3 outbox worker → SMTP; reminder or schedule-change item.	Worker claims persisted item; current source has configurable attempts/delays and provider outcomes. Ambiguous send can be uncertain.	No live send, receipt, bounce, credentials, or provider outcome verified. Reconcile ambiguity; do not blindly resend.
M3 Google Calendar	M3 adapter/worker → Google Calendar for session create/update/cancel.	Outbox/adapter result handling is source-visible; authenticated client absent in reviewed evidence.	No live event operation or authenticated client verification.
VNPay	M2 deposit/M5 payment URL and callback paths ↔ VNPay.	Source/config and callback handling exist; payment success and stay closure are distinct. Runtime/idempotency details remain partial.	VNPay is the only in-scope method by Revision 1.1; no live gateway, settlement, or finance reconciliation evidence.
Schedulers / outbox / webhook	Scheduled M3 workers and payment callback handlers → database/provider boundaries.	30-second dispatch sweep and 60-second recovery schedule are source declarations; operation depends on activation/configuration.	No activation, provider delivery, shutdown, or production timing verified.
VietQR source	Legacy/config or source endpoints may remain; not an approved payment method in current scope.	Source presence does not establish active operational scope.	Deferred/out of current strategic payment scope; no provider or settlement verification.

5. Deployment, SQL Server and migration status

Figure 18 — Intended deployment relationships — topology unknown

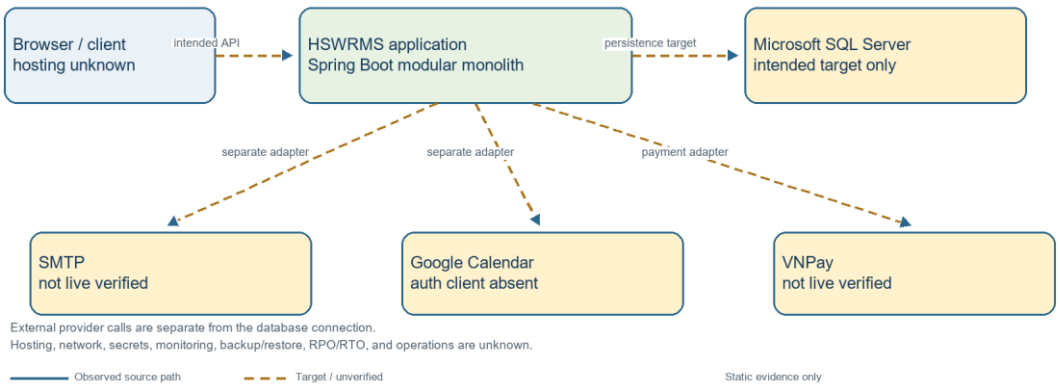


Figure 25. Intended deployment relationships. Hosting, topology, SQL Server, and provider operations remain unverified.

No approved topology, network segmentation, secret handling, health checks, logging/monitoring, backup/restore, RPO/RTO, certificate rotation, scaling or operations ownership was available. H2 is test evidence only and does not establish SQL Server behavior.

Figure 25 separates intended browser-to-application, application-to-database, and application-to-provider relationships. External providers are not shown as reached through SQL Server.

Static migration history includes duplicate V1_6 identifiers; V004/V2 namespace ambiguity; overlapping V003/V1_2/V1_3/V1_4 ownership; V011/V012/V015 M3 consolidation; V014/V017

trigger/payment overlap; destructive V1_5; and seed-like DML in V007. Filenames do not define an approved order. No migration ran and this SDS does not select an execution plan.

Migration question	SDS treatment	Evidence required
Duplicate V1_6; V004 vs V2	No order inferred.	DBA-approved framework and unique ordered manifest.
V003 / V1_2 / V1_3 / V1_4 overlap	Potential duplicate/conflicting object ownership.	Baseline-to-file DDL diff and object-owner confirmation.
V011 / V012 / V015	Potential M3 schema consolidation.	Dependency and data impact review plus SQL Server restore test.
V014 vs V017	Payment/type/VietQR vs trigger/payment overlap.	Compare against canonical schema, trigger definitions and approved finance contract.
V1_5 destructive removal	Potential loss of duration-unit data.	Impact assessment and explicit approval before execution.
V007 seed-like DML	Do not treat seed content as schema migration.	Approved seed lifecycle and separated deployment artifact.

6. Requirements, API, service, data, and test traceability

Phase 0.1-F reports 46 FR records: 19 have partial direct controller/dependency candidates; 27 are not defensibly mapped; 0 complete FR-to-acceptance chains. The table reproduces each FR ID, summary and candidate API/status from A01. “Candidate” is not a verified implementation chain. Full records remain in SDS_REPOSITORY_AUDIT_PHASE_0_1_F_TECHNICAL_EVIDENCE/A01_FR_IMPLEMENTATION_TRACEABILITY.csv.

FR ID	Requirement	Module	API candidate	Evidence status
FR-M1-01	Register Account	M1	/api/v1/auth/register	Partial candidate
FR-M1-02	Email OTP Verification	M1	/api/v1/auth/verify-email	Partial candidate
FR-M1-03	Email & Password Login	M1	/api/v1/auth/login	Partial candidate
FR-M1-04	Forgot & Reset Password	M1	/api/v1/auth/forgot-password;/api/v1/auth/reset-password	Partial candidate
FR-M1-05	Single Sign-On Authentication (Google & Facebook)	M1	/api/v1/auth/register/sso;/api/v1/auth/sso/tickets/exchange;/api/v1/auth/sso/tickets/resolve;/api/v1/auth/sso/tickets/link	Partial candidate
FR-M1-06	Account Settings Management	M1	UNMAPPED	Not mapped
FR-M1-07	Health & Dietary Profile Management	M1	UNMAPPED	Not mapped

FR ID	Requirement	Module	API candidate	Evidence status
FR-M1-08	Consent Management	M1	/api/v1/customers/me/consents/{consentType}/revoke	Partial candidate
FR-M1-09	Health Data Deletion Request & Retention Processing	M1	UNMAPPED	Not mapped
FR-M2-01	Browse Retreat Packages	M2	UNMAPPED	Not mapped
FR-M2-02	Book Retreat & Pay Deposit	M2	UNMAPPED	Not mapped
FR-M2-03	Perform Check-in & Assign Villa	M2	UNMAPPED	Not mapped
FR-M2-04	Manage Villa Status	M2	/api/v1/villas/{villaNumber}/status	Partial candidate
FR-M2-05	View Booking & Itinerary	M2	/api/v1/customer/bookings/{id}	Partial candidate
FR-M2-06	Create Walk-in Booking	M2	/api/v1/receptionist/walk-in	Partial candidate
FR-M2-07	Cancel Booking or Services	M2	/api/v1/customer/bookings/{id}/cancel	Partial candidate
FR-M2-08	Modify Booking or Services	M2	/api/v1/customer/bookings/{id}	Partial candidate
FR-M3-01	Submit Spa/Therapy Scheduling Request	M3	/api/v1/spa/sessions	Partial candidate
FR-M3-02	Generate & Select Valid Scheduling Candidates	M3	/api/v1/spa/candidates	Partial candidate

FR ID	Requirement	Module	API candidate	Evidence status
FR-M3-03	Revalidate & Atomically Reserve Selected Schedule	M3	/api/v1/spa/sessions	Partial candidate
FR-M3-04	Manage Reminder Timeline & Customer Confirmation	M3	/api/v1/spa/sessions/{id}/confirm-attendance	Partial candidate
FR-M3-05	Recover / Re-Match Disrupted Session	M3	/api/v1/spa/sessions/{id}/decline-attendance	Partial candidate
FR-M3-06	Receptionist-Initiated Extra Spa Booking	M3	UNMAPPED	Not mapped
FR-M3-07	View Assigned Schedule & Treatment Safety Health Profile	M3	UNMAPPED	Not mapped
FR-M3-08	Record Final Session Attendance & Outcome	M3	/api/v1/spa/sessions/{id}/confirm-attendance	Partial candidate
FR-M4-01	View Personalized Menu	M4	UNMAPPED	Not mapped
FR--M4-02	Pre-Select Daily Meal	M4	UNMAPPED	Not mapped
FR--M4-03	Order Extra A-lar-carte	M4	/api/v1/fnb/customer/daily-meals;/api/v1/fnb/orders/daily-meals	Partial candidate
FR--M4-04	Manage Ingredient Batch Information	M4	UNMAPPED	Not mapped
FR--M4-05	View Meal Preparation Dashboard	M4	UNMAPPED	Not mapped
FR--M4-06	Prepare Meal	M4	UNMAPPED	Not mapped

FR ID	Requirement	Module	API candidate	Evidence status
FR-	[FR-] Update Meal Order Status 5. Module 5 - Consolidated Checkout & Analytics	UNRESOLVED	UNMAPPED	Not mapped
FR--M5-01	Generate Consolidated Invoice & Manage Folio	M5	UNMAPPED	Not mapped
FR--M5-02	Process Final Payment & Settlement	M5	UNMAPPED	Not mapped
FR--M5-03	Finalize Stay & Automated Checkout	M5	/api/v1/billing/invoices/{invoiceId}/finalize-checkout	Partial candidate
FR-M5-04	View & Download E-Invoice	M5	UNMAPPED	Not mapped
FR-M5-05	Submit Customer Review & Rating	M5	UNMAPPED	Not mapped
FR-M5-06	View Operational & Revenue Analytics Dashboard	M5	UNMAPPED	Not mapped
FR-M6-01	Manage Staff Accounts & Role Assignment	M6	UNMAPPED	Not mapped
FR-M6-02	Manage Master Data	M6	UNMAPPED	Not mapped
FR-M6-03	Configure Business Parameters	M6	UNMAPPED	Not mapped
FR-M6-04	Toggle Feature Flags	M6	UNMAPPED	Not mapped

FR ID	Requirement	Module	API candidate	Evidence status
FR-M6-05	Configure Permission Matrix	M6	UNMAPPED	Not mapped
FR-M6-06	Delegate Limited Admin Rights to Manager	M6	UNMAPPED	Not mapped
FR-M6-07	View Sensitive-Data Audit Log	M6	UNMAPPED	Not mapped
FR-M6-08	View System Overview Dashboard	M6	UNMAPPED	Not mapped

Original FR identifiers are retained exactly as recorded in A01, including malformed or unresolved source identifiers; no IDs were renumbered.

Trace-link statuses: route and call-site matches are direct static evidence, not requirement satisfaction; unresolved service/data joins are partial; test-method associations without approved FR linkage remain inferred; unmapped/conflicting identifiers stay raw. Phase 0.1-F records 0 direct approved test-to-FR/rule links, 0 tests run in that phase, and 0 complete FR-to-acceptance chains. Raw "FR-" remains unresolved.

Test-source evidence summary

Source group	Test classes	Annotated methods	Direct FR/rule link
SHARED	13	79	0 direct; inferred links only
m1identity	24	148	0 direct; inferred links only
m2booking	8	58	0 direct; inferred links only
m3spa	17	154	0 direct; inferred links only
m4fnb	11	100	0 direct; inferred links only
m5billing	6	135	0 direct; inferred links only
m6admin	9	63	0 direct; inferred links only

Additional static crosswalks: B01–B05 route/call/auth/error; C01–C04 repository/entity/table/ownership; D01–D04 module boundaries; E01–E05 cross-module/failure; F01–F04 test source and acceptance gaps; G01–G05 architecture/data/state facts; H01–H02 technical gaps. No tests were run for this SDS.

7. Strategic decision register: Revision 1.1

Revision 1.1 (01 November 2026) is the latest user-confirmed strategic decision baseline used here. Its decisions are target direction, not implementation facts or formal organizational, finance, privacy, security, legal or operations approvals. Cross-check against Phase 0.1-E/F showed the following remaining gaps.

Decision	Strategic direction in Rev 1.1	Implementation/evidence reconciliation	Status
PRJ-SRS	Option D: authoritative SRS baseline unresolved.	Phase 0.1-F also reports source authority unresolved; no version is called approved.	OPEN

Decision	Strategic direction in Rev 1.1	Implementation/evidence reconciliation	Status
OI-01	M5 owns idempotent Folio posting contract.	M3 adapter still uses non-idempotent postFolioCharge and catches/logs; M4 has idempotent source key. Risk not remediated.	TARGET / GAP
OI-02	Option B: one initial attempt plus four retries (five total), delayed 30s, 1m, 2m, 4m; reconcile ambiguous SMTP outcomes, do not blindly resend.	Prior decision register records this policy, but historical ADR-06 status remains open; implementation/provider semantics not fully verified.	PARTIAL / ADR CONFLICT
OI-03	Staged rollout: Email → Bell/In-app → Push.	Email required; staged rollout is target only; later channels/runtime not verified.	TARGET
OI-04	M5 owns PaymentTransaction; M2 owns Booking and deposit rules.	Code adjacency does not prove enforcement; transaction boundary and reconciliation remain to be mapped.	TARGET / PARTIAL
OI-05	VNPay only; defer Cash/POS and other methods.	VNPay code exists; no live gateway. Do not represent other methods as current scope.	TARGET / RUNTIME GAP
OI-06	Explicit configurable service-location contract.	Source/config ownership and field/value contract need mapping.	TARGET / PARTIAL
OI-07	Package-specific entitlement matrix with effective dates.	No complete implementation/configuration evidence in Phase 0.1-F.	TARGET / GAP
OI-08	Persisted IN_PROGRESS prohibited.	ADR-07 and M3 state evidence retain the prohibition.	DIRECTION / SOURCE BACKED
OI-09	Confirmation lifecycle remains separate.	Current confirmation_status is separate; historical ADR reconciliation still required.	SOURCE / ADR CONFLICT
M1	Consent, retention, minimum access; values/semantics not invented.	Project rules align; detailed retention/access matrix and org/privacy approval missing.	DIRECTION / OPEN VALUES
M2	M2 owns Booking/deposit business rules.	Target ownership; verify all payment/deposit paths.	TARGET
M3	Cross-customer recovery/swap needs explicit mutual consent.	Implementation report says workflow remains partial.	TARGET / GAP
M4	Allergy suitability fails closed.	Project rule and source path exist; full runtime/edge coverage not demonstrated.	TARGET / PARTIAL
M5	M5 owns PaymentTransaction/idempotent Folio contract; VNPay only.	Adapter risk and live gateway limitations remain.	TARGET / GAP
M6	Delegation least privilege, time-limited, auditable.	Selected create/revoke source paths exist; full consumers and formal security approval not established.	TARGET / PARTIAL

Revision 1.1 expressly does not supersede technical evidence or resolve SRS authority. Where historical ADRs differ, this SDS records the conflict and does not rewrite repository history.

8. Gaps and required evidence

Gap	Current status	Evidence/action required
SRS authority and formal approvals	OPEN / NOT PROVIDED	Approve authoritative SRS version; record organization owner and signed/versioned approval separately from user strategic direction.
OI-01 implementation risk	OPEN — code mismatch	Update all M3/M4 posting paths to M5 idempotent contract only under implementation authorization; verify failure/retry/reconciliation and source identity.

Gap	Current status	Evidence/action required
OI-02 / OI-09 historical ADRs	CONFLICTING	Record superseding ADR/history; do not silently edit status.
M1 privacy and retention	PARTIAL	Privacy/security-approved data access, consent, retention and deletion matrix; no values inferred.
M2 configuration and ownership	PARTIAL	Booking/deposit, guest-count, zone/villa and package rule mapping plus tests and approval.
M3 mutual consent and runtime	PARTIAL / NOT VERIFIED	Mutual-consent workflow and evidence; SQL Server concurrency and provider behavior.
M4 allergy/location/entitlements	PARTIAL / NOT VERIFIED	Fail-closed tests and explicit service-location/package entitlement configuration.
M5 payments/refunds/SQL Server	PARTIAL / NOT VERIFIED	VNPay sandbox evidence; finance-owned refund/reconciliation decisions; SQL Server trigger/transaction check.
M6 contracts and consumers	PARTIAL	Canonical SystemConfiguration/RightDelegation contract; all protected endpoints/delegation consumers; security approval.
Database/migration history	NOT VERIFIED	DBA catalog, resolved baseline/sequence, test-copy restore and rollback evidence.
SMTP / Calendar / deployment	NOT VERIFIED	SMTP sandbox, authenticated Calendar client test, approved deployment/operations evidence.
Requirement/test trace	INCOMPLETE	Join each FR to authoritative use case/API/service/data/test and acceptance record. Phase 0.1-F has 0 complete chains.

Evidence index

Existing V2: 06_Reports/HSWRMS_Software_Design_Specification_V2_Final.docx and .pdf. Companion registers: SDS_SOURCE_TRACEABILITY_FINAL.md; SDS_REQUIREMENT_TRACEABILITY_FINAL.md; SDS_DATABASE_DICTIONARY_FINAL.md; SDS_API_INVENTORY_FINAL.md; SDS_MIGRATION_ANALYSIS_FINAL.md; SDS_DECISION_REGISTER_FINAL.md; SDS_DOCUMENTATION_GAP_REGISTER_FINAL.md; SDS_FINAL_QA_REPORT.md.

Phase 0.1-F package: SDS_REPOSITORY_AUDIT_PHASE_0_1_F_TECHNICAL_EVIDENCE/PHASE_0_1_F_FINAL_REPORT.md and registers A01–H02. Strategic baseline: Human Decision Disposition Revision 1.1 (HSWRMS_Human_Decision_Disposition_Revision_1.1.docx) (decision date 01 November 2026).

9. Prior QA evidence and final status

The existing V2 Final QA report states 19 PDF pages were rendered and visually reviewed, with 11 embedded PNG diagrams; it describes them as simplified, not complete class/ER/sequence models. Prior reports record M3 focused H2 results of 129 passed, full backend H2 results of 770 passed and 10 skipped, and 30/30 historical API scenarios. These are report-reported prior results, not rerun here. Phase 0.1-F says 0 tests ran in that pass. This final SDS task ran no tests, build, application, migration, DB operation or provider call.

Area	Status	Limit
Repository design	PARTIAL	Static source trace only; 27 FRs unmapped; 0 complete FR acceptance chains.
Requirement authority	NOT APPROVED	PRJ-SRS Option D remains unresolved.
SQL Server/migrations	NOT VERIFIED	No connection or migration; H2 does not prove SQL Server.

Area	Status	Limit
SMTP/Calendar/VNPay/Push	NOT VERIFIED	No live provider operation; Calendar client absent.
Deployment/operations	NOT VERIFIED	No approved topology, secrets/monitoring/restore plan.
Organizational acceptance/G0	NOT PROVIDED	Revision 1.1 is user strategic direction only; no organizational signoff or G0 acceptance.

Conclusion: This repository-grounded specification separates static source facts, user-approved strategic direction, proposed contracts, conflicts, and unverified runtime or approval evidence. It is not organizationally approved or accepted. Every PARTIAL / NOT VERIFIED / OPEN state remains until named evidence is supplied.